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RINVOQ®15 mg extended-release film coated tablets
RINVOQ®30 mg extended-release film coated tablets
RINVOQ®45 mg extended-release film coated tablets

QUALITATIVE AND QUANTITATIVE COMPOSITION

RINVOQ® 15 mg extended-release film coated tablets
Each extended-release tablet contains upadacitinib hemihydrate, equivalent to 15 mg of upadacitinib.

RINVOQ® 30 mg extended-release film coated tablets
Each prolonged-release tablet contains upadacitinib hemihydrate, equivalent to 30 mg of upadacitinib.

RINVOQ® 45 mg extended-release film coated tablets
Each extended-release tablet contains upadacitinib hemihydrate, equivalent to 45 mg of upadacitinib.

For the full list of excipients, see **List of excipients** section.

PHARMACEUTICAL FORM

Extended-release tablet.

RINVOQ® 15 mg extended-release film coated tablets
Purple 14 x 8 mm, oblong biconvex extended-release tablets imprinted on one side with ‘a15’.

RINVOQ® 30 mg extended-release film coated tablets
Red 14 x 8 mm, oblong biconvex prolonged-release tablets imprinted on one side with ‘a30’.

RINVOQ® 45 mg extended-release film coated tablets
Yellow biconvex oblong, with dimensions of 14 x 8 mm, and debossed with ‘a45’ on one side.

CLINICAL PARTICULARS

Therapeutic indications

For RINVOQ® 15 mg only:

<u>Rheumatoid arthritis</u>

RINVOQ® is indicated for the treatment of moderate to severe active rheumatoid arthritis in adult patients who have responded inadequately to, or who are intolerant to one or more disease-modifying anti-rheumatic drugs (DMARDs). RINVOQ® may be used as monotherapy or in combination with methotrexate.

<u>Psoriatic arthritis</u>

RINVOQ® is indicated for the treatment of active psoriatic arthritis in adult patients who have responded inadequately to, or who are intolerant to one or more DMARDs. RINVOQ® may be used as monotherapy or in combination with methotrexate.

Non-radiographic axial spondyloarthritis (nr-axSpA)

RINVOQ® is indicated for the treatment of active non-radiographic axial spondyloarthritis in adult patients with objective signs of inflammation as indicated by elevated C-reactive protein (CRP) and/or magnetic resonance imaging (MRI), who have responded inadequately to nonsteroidal anti-inflammatory drugs (NSAIDs).

Ankylosing Spondylitis (AS, radiographic axial spondyloarthritis)

RINVOQ® is indicated for the treatment of active ankylosing spondylitis in adult patients who have responded inadequately to conventional therapy.

Giant cell arteritis

RINVOQ® is indicated for the treatment of giant cell arteritis in adult patients.

For RINVOQ® 15 mg and 30 mg:

Atopic dermatitis

RINVOQ® is indicated for the treatment of moderate to severe atopic dermatitis in adults and adolescents 12 years and older who are candidates for systemic therapy.

For RINVOQ® 15 mg, 30 mg and 45 mg:

Ulcerative Colitis

RINVOQ® is indicated for the treatment of adult patients with moderately to severely active ulcerative colitis who have had an inadequate response, lost response or were intolerant to either conventional therapy or a biologic agent.

Crohn's disease

RINVOQ® is indicated for the treatment of adult patients with moderately to severely active Crohn's disease who have had an inadequate response, lost response or were intolerant to either conventional therapy or a biologic agent.

Posology and method of administration

Treatment with upadacitinib should be initiated and supervised by physicians experienced in the diagnosis and treatment of conditions for which upadacitinib is indicated.

Posology

Rheumatoid arthritis, psoriatic arthritis, non-radiographic axial spondyloarthritis and ankylosing spondylitis

The recommended dose of upadacitinib is 15 mg once daily.

Consideration should be given to discontinuing treatment in patients with non-radiographic axial spondyloarthritis and ankylosing spondylitis who have shown no clinical response after 16 weeks of treatment. Some patients with initial partial response may subsequently improve with continued treatment beyond 16 weeks.

Giant cell arteritis

The recommended dose of upadacitinib is 15 mg once daily in combination with a tapering course of corticosteroids. Upadacitinib monotherapy should not be used for the treatment of acute relapses (see section **Special warnings and precautions for use**).

Based upon the chronic nature of giant cell arteritis, upadacitinib 15 mg once daily can be continued as monotherapy following discontinuation of corticosteroids. Treatment beyond 52 weeks should be guided by disease activity, physician discretion, and patient choice.

Atopic dermatitis

Adults and adolescent patients 12 years of age and older weighing at least 40 kg

The recommended dose of RINVOQ is 15 mg once daily.

- A dose of 30 mg once daily may be appropriate for some patients, such as those with high disease burden.
- A dose of 30 mg once daily may be appropriate for patients who do not show adequate therapeutic benefit to 15 mg once daily.
- The lowest effective dose for maintenance should be used.

For patients ≥ 65 years of age, the recommended dose of RINVOQ is 15 mg once daily.

RINVOQ has not been studied in adolescents weighing less than 40 kg.

Concomitant topical therapies
Upadacitinib can be used with or without topical corticosteroids. Topical calcineurin inhibitors may be used for sensitive areas such as the face, neck, and intertriginous and genital areas.

Ulcerative Colitis

Induction
The recommended induction dose of RINVOQ is 45 mg once daily for 8 weeks. For patients who do not achieve adequate therapeutic benefit by week 8, RINVOQ 45 mg once daily may be continued for an additional 8 weeks (see **Undesirable effects** and **Pharmacodynamic properties** sections). RINVOQ should be discontinued in any patient who shows no evidence of therapeutic benefit by week 16.

Maintenance
The recommended maintenance dose of RINVOQ is 15 mg or 30 mg once daily based on individual patient presentation:

- A dose of 30 mg once daily may be appropriate for some patients, such as those with high disease burden or requiring 16-week induction treatment.
- A dose of 30 mg once daily may be appropriate for patients who do not show adequate therapeutic benefit to 15 mg once daily.
- The lowest effective dose for maintenance should be used.

For patients ≥ 65 years of age, the recommended maintenance dose is 15 mg once daily.

In patients who have responded to treatment with RINVOQ, corticosteroids may be reduced and/or discontinued in accordance with standard of care.

Crohn's disease

Induction

The recommended induction dose of RINVOQ is 45 mg once daily for 12 weeks. For patients who have not achieved adequate therapeutic benefit after the initial 12-week induction, prolonged induction for an additional 12 weeks with a dose of 30 mg once daily may be considered. For these patients, RINVOQ should be discontinued if there is no evidence of therapeutic benefit after 24 weeks of treatment.

Maintenance

The recommended maintenance dose of RINVOQ is 15 mg or 30 mg once daily based on individual patient presentation:

- A dose of 30 mg once daily may be appropriate for patients with high disease burden or those who do not show adequate therapeutic benefit with 15 mg once daily.
- The lowest effective dose for maintenance should be used.

For patients ≥ 65 years of age, the recommended maintenance dose is 15 mg once daily.

In patients who have responded to treatment with RINVOQ, corticosteroids may be reduced and/or discontinued in accordance with standard of care.

Interactions

For patients with ulcerative colitis and Crohn's disease receiving strong inhibitors of cytochrome P450 (CYP) 3A4 (e.g., ketoconazole, clarithromycin), the recommended induction dose is 30 mg once daily and the recommended maintenance dose is 15 mg once daily (see **Interaction with other medicinal products and other forms of interaction**).

Dose initiation

Treatment should not be initiated in patients with an absolute lymphocyte count (ALC) that is < 0.5 x 10^9 cells/L, an absolute neutrophil count (ANC) that is < 1 x 10^9 cells/L or who have haemoglobin (Hb) levels that are < 8 g/dL (see **Special warnings and precautions and undesirable effects** section).

Dose Interruption

Treatment should be interrupted if a patient develops a serious infection until the infection is controlled.

Interruption of dosing may be needed for management of laboratory abnormalities as described in Table 1.

Table 1. Laboratory measures and monitoring guidance

Laboratory measure	Action	Monitoring guidance
Absolute Neutrophil Count (ANC)	Treatment should be interrupted if ANC is < 1 x 10^9 cells/L and may be restarted once ANC returns above this value	Evaluate at baseline and then no later than 12 weeks after initiation of treatment. Thereafter evaluate according to individual patient management
Absolute Lymphocyte Count (ALC)	Treatment should be interrupted if ALC is < 0.5 x 10^9 cells/L and may be restarted once ALC returns above this value	
Haemoglobin (Hb)	Treatment should be interrupted if Hb is < 8 g/dL and may be restarted once Hb returns above this value	
Hepatic transaminases	Treatment should be temporarily interrupted if drug-induced liver injury is suspected	Evaluate at baseline and thereafter according to routine patient management.
Lipids	Patients should be managed according to international clinical guidelines for hyperlipidaemia	12 weeks after initiation of treatment and thereafter according to international clinical guidelines for hyperlipidaemia

<u>Special Populations</u>

Elderly

Rheumatoid arthritis, psoriatic arthritis, non-radiographic axial spondyloarthritis and ankylosing spondylitis
There are limited data in patients aged 75 years and older.

Atopic dermatitis
For atopic dermatitis, doses higher than 15 mg once daily are not recommended in patients aged 65 years and older (see **undesirable effects** section).

Ulcerative colitis and Crohn's disease
For ulcerative colitis and Crohn's disease, doses higher than 15 mg once daily for maintenance therapy are not recommended in patients 65 years of age and older (see **undesirable effects** section). The safety and efficacy of upadacitinib in patients 75 years of age and older have not yet been established.

Renal Impairment

No dose adjustment is required in patients with mild or moderate renal impairment. There are limited data on the use of upadacitinib in subjects with severe renal impairment (see **Pharmacokinetic properties** section). Upadacitinib should be used with caution in patients with severe renal impairment. The use of upadacitinib has not been studied in subjects with end stage renal disease and is therefore not recommended for use in these patients.

For patients with severe renal impairment, the following dose adjustments are recommended:

Table 2. Recommended Dose for Severe Renal Impairment [a]

	Indication	Recommended once daily dose
Severe renal impairment	Rheumatoid arthritis, psoriatic arthritis, non-radiographic axial spondyloarthritis, ankylosing spondylitis, giant cell arteritis, atopic dermatitis	15 mg
	Ulcerative Colitis, Crohn's disease	Induction: 30 mg
		Maintenance: 15 mg
[a]estimated glomerular filtration rate (eGFR) 15 to < 30 ml/min/1.73m^2		

Hepatic Impairment

No dose adjustment is required in patients with mild (Child Pugh A) or moderate (Child Pugh B) hepatic impairment (see **Pharmacokinetic properties** section). Upadacitinib should not be used in patients with severe (Child Pugh C) hepatic impairment (see **Contraindications** section).

Paediatric population

The safety and efficacy of RINVOQ® in adolescents weighing < 40 kg and in children aged 0 to less than 12 years have not yet been established. No data are available.

The safety and efficacy of RINVOQ® in children and adolescents with rheumatoid arthritis, psoriatic arthritis, non-radiographic axial spondyloarthritis, ankylosing spondylitis, ulcerative colitis and Crohn's disease aged 0 to less than 18 years have not yet been established. No data are available.

There is no relevant use of RINVOQ® in the paediatric population in the indication giant cell arteritis.

Method of administration

RINVOQ® is to be taken orally once daily with or without food and may be taken at any time of the day. Tablets should be swallowed whole and should not be split, crushed, or chewed in order to ensure the entire dose is delivered correctly.

Missed Dose

If a dose of RINVOQ® is missed and it is more than 10 hours from the next scheduled dose, advise the patient to take a dose as soon as possible and then to take the next dose at the usual time. If a dose is missed and it is less than 10 hours from the next scheduled dose, advise the patient to skip the missed dose and take only a single dose as usual the following day. Advise the patient not to double a dose to make up for a missed dose.

Contraindications

- Hypersensitivity to the active substance or to any of the excipients listed in the **List of excipients** section.
- Active tuberculosis (TB) or active serious infections (see **Special warnings and precautions for use** section).
- Severe hepatic impairment (see **Posology and method of administration** section).
- Pregnancy (see **Fertility, pregnancy and lactation** section).

Special warnings and precautions for use

Immunosuppressive Medicinal Products

Combination with other potent immunosuppressants such as azathioprine, 6-mercaptopurine, ciclosporin, tacrolimus, and biologic DMARDs or other Janus kinase (JAK) inhibitors has not been evaluated in clinical studies and is not recommended as a risk of additive immunosuppression cannot be excluded.

Serious Infections

Serious and sometimes fatal infections have been reported in patients receiving upadacitinib. The most frequent serious infections reported with upadacitinib included pneumonia and cellulitis (see **Undesirable effects** section). Cases of bacterial meningitis have been reported in patients receiving upadacitinib. Among opportunistic infections, tuberculosis, multidermatomal herpes zoster, oral/oesophageal candidiasis, and cryptococcosis were reported with upadacitinib. A higher rate of serious infections was observed with upadacitinib 30 mg compared to upadacitinib 15 mg.

Upadacitinib should not be initiated in patients with an active, serious infection, including localised infections.

Consider the risks and benefits of treatment prior to initiating upadacitinib in patients:

- with chronic or recurrent infection
- who have been exposed to tuberculosis
- with a history of a serious or an opportunistic infection
- who have resided or travelled in areas of endemic tuberculosis or endemic mycoses; or
- with underlying conditions that may predispose them to infection.

Patients should be closely monitored for the development of signs and symptoms of infection during and after treatment with upadacitinib. Upadacitinib therapy should be interrupted if a patient develops a serious or opportunistic infection. A patient who develops a new infection during treatment with upadacitinib should undergo prompt and complete diagnostic testing appropriate for an immunocompromised patient; appropriate antimicrobial therapy should be initiated, the patient should be closely monitored, and upadacitinib therapy should be interrupted if the patient is not responding to antimicrobial therapy. Upadacitinib therapy may be resumed once the infection is controlled.

As there is a higher incidence of infections in the elderly ≥ 65 years of age, caution should be used when treating this population.

Tuberculosis

Patients should be screened for tuberculosis (TB) before starting upadacitinib therapy. Upadacitinib should not be given to patients with active TB (see **Contraindications** section). Anti-TB therapy should be considered prior to initiation of upadacitinib in patients with previously untreated latent TB or in patients with risk factors for TB infection.

Consultation with a physician with expertise in the treatment of TB is recommended to aid in the decision about whether initiating anti-TB therapy is appropriate for an individual patient.

Patients should be monitored for the development of signs and symptoms of TB, including patients who tested negative for latent TB infection prior to initiating therapy.

Viral Reactivation

Viral reactivation, including cases of herpes virus reactivation (e.g., herpes zoster), was reported in clinical studies (see **Undesirable effects** section). The risk of herpes zoster appears to be higher in patients treated with RINVOQ® in Japan. If a patient develops herpes zoster, interruption of upadacitinib therapy should be considered until the episode resolves.

Screening for viral hepatitis and monitoring for reactivation should be performed before starting and during therapy with upadacitinib. Patients who were positive for hepatitis C antibody and hepatitis C virus RNA were excluded from clinical studies. Patients who were positive for hepatitis B surface antigen or hepatitis B virus DNA were excluded from clinical studies. If hepatitis B virus DNA is detected while receiving upadacitinib, a liver specialist should be consulted.

Vaccination

No data are available on the response to vaccination with live or inactivated vaccines in patients receiving upadacitinib. Use of live, attenuated vaccines during or immediately prior to upadacitinib therapy is not recommended. Prior to and during upadacitinib treatment, it is recommended that patients be brought up to date with all immunisations, including prophylactic zoster vaccinations, in agreement with current immunisation guidelines. (*see* **PHARMACOLOGICAL PROPERTIES**).

Malignancy

The risk of malignancies, including lymphoma is increased in patients with rheumatoid arthritis. Immunomodulatory medicinal products may increase the risk of malignancies, including lymphoma. The clinical data are currently limited and long-term studies are ongoing.
In a large randomized active-controlled study in rheumatoid arthritis patients 50 years and older with at least one additional cardiovascular risk factor, an increased incidence of malignancy, particularly lung cancer, lymphoma and non-melanoma skin cancer [NMSC], was observed with tofacitinib (a different JAK inhibitor) compared to Tumor Necrosis Factor (TNF) blockers. The higher rate of malignancy was primarily observed in patients 65 years of age and older and patients who are current or past long-time smokers.

Malignancies were observed in clinical studies of upadacitinib. A higher rate of malignancies, driven by NMSC, was observed with RINVOQ® 30 mg compared to RINVOQ® 15 mg. The risks and benefits of upadacitinib treatment should be considered prior to initiating therapy in patients with a known malignancy other than a successfully treated NMSC or when considering continuing upadacitinib therapy in patients who develop a malignancy.

Non-melanoma skin cancer

NMSCs have been reported in patients treated with upadacitinib. Periodic skin examination is recommended for patients who are at increased risk for skin cancer.

Haematological abnormalities

Absolute Neutrophil Count (ANC) < 1 x 10^9 cells/L, Absolute Lymphocyte Count (ALC) < 0.5 x 10^9 cells/L and haemoglobin < 8 g/dL were reported in ≤1 % of patients in clinical trials (see **Undesirable effects** section). Treatment should not be initiated, or should be temporarily interrupted, in patients with an ANC < 1 x 10^9 cells/L, ALC < 0.5 x 10^9 cells/L or haemoglobin < 8 g/dL observed during routine patient management (see **Posology and method of administration** section).

Major Adverse Cardiovascular Events (MACE)

In a large randomized active-controlled study in rheumatoid arthritis patients 50 years and older with at least one additional cardiovascular risk factor, an increased incidence of MACE, including myocardial infarction (MI), was observed with tofacitinib (a different JAK inhibitor) compared with TNF blockers. The higher rate of MACE was primarily observed in patients 65 years of age and older, patients with a history of atherosclerotic cardiovascular disease, and patients with other cardiovascular risk factors (such as current or past long-time smokers).

Consider the risks and benefits of RINVOQ® treatment prior to initiating therapy in patients with cardiovascular risk factors or when considering continuing RINVOQ® in patients who develop MACE.

Lipids

Treatment with upadacitinib was associated with dose-dependent increases in lipid parameters, including total cholesterol, low-density lipoprotein (LDL) cholesterol, and high-density lipoprotein (HDL) cholesterol (see **Undesirable effects** section). Elevations in LDL cholesterol decreased to pre-treatment levels in response to statin therapy, although evidence is limited. The effect of these lipid parameter elevations on cardiovascular morbidity and mortality has not been determined (see **Posology and method of administration** section for monitoring guidance).

Hepatic transaminase elevations

Treatment with upadacitinib was associated with an increased incidence of liver enzyme elevation compared to placebo.

Evaluate at baseline and thereafter according to routine patient management. Prompt investigation of the cause of liver enzyme elevation is recommended to identify potential cases of drug-induced liver injury.

If increases in ALT or AST are observed during routine patient management and drug-induced liver injury is suspected, upadacitinib therapy should be interrupted until this diagnosis is excluded.

Venous Thromboembolism

Events of deep venous thrombosis (DVT) and pulmonary embolism (PE) have been reported in patients receiving JAK inhibitors including upadacitinib. In a large randomized active-controlled study in rheumatoid arthritis patients 50 years and older with at least one additional cardiovascular risk factor, a dose-dependent increased incidence of VTE was observed with tofacitinib (a different JAK inhibitor) compared with TNF blockers.
Upadacitinib should be used with caution in patients at high risk for DVT/PE. Risk factors that should be considered in determining the patient's risk for DVT/PE include older age, obesity, a medical history of DVT/PE, patients undergoing major surgery, and prolonged immobilisation. If clinical features of DVT/PE occur, upadacitinib treatment should be discontinued and patients should be evaluated promptly, followed by appropriate treatment.

Hypersensitivity Reactions

Serious hypersensitivity reactions such as anaphylaxis and angioedema were reported in patients receiving RINVOQ® in clinical trials. If a clinically significant hypersensitivity reaction occurs, discontinue RINVOQ® and institute appropriate therapy (*see* **Undesirable effects** section).

Gastrointestinal Perforations

Events of gastrointestinal perforations have been reported in clinical trials (*see* **ADVERSE REACTIONS**) and from post-marketing sources. RINVOQ® should be used with caution in patients who may be at risk for gastrointestinal perforation (e.g., patients with diverticular disease, a history of diverticulitis, or who are taking nonsteroidal anti-inflammatory drugs (NSAIDs), corticosteroids, or opioids). Patients presenting with new onset abdominal signs and symptoms should be evaluated promptly for early identification of gastrointestinal perforation.

Medication Residue in Stool

Reports of medication residue in stool or ostomy output have occurred in patients taking RINVOQ. Most reports described anatomic (e.g., ileostomy, colostomy, intestinal resection) or functional gastrointestinal conditions with shortened gastrointestinal transit times. Patients should be instructed to contact their healthcare provider if medication residue is observed repeatedly. Patients should be clinically monitored, and alternative treatment should be considered if there is an inadequate therapeutic response.

Giant Cell Arteritis

Upadacitinib monotherapy should not be used for the treatment of acute relapses as efficacy in this setting has not been established. Corticosteroids should be given according to medical judgement and practice guidelines.

Interaction with other medicinal products and other forms of interaction

Potential for other medicinal products to affect the pharmacokinetics of upadacitinib

Upadacitinib is metabolised mainly by CYP3A4. Therefore, upadacitinib plasma exposures can be affected by medicinal products that strongly inhibit or induce CYP3A4.

Coadministration with CYP3A4 Inhibitors

Upadacitinib exposure is increased when co-administered with strong CYP3A4 inhibitors (such as ketoconazole, itraconazole, posaconazole, voriconazole, and clarithromycin and grapefruit). In a clinical study, coadministration of upadacitinib with ketoconazole resulted in 70% and 75% increases in upadacitinib C_{max} and AUC, respectively. Upadacitinib 15 mg once daily should be used with caution in patients receiving chronic treatment with strong CYP3A4 inhibitors. Upadacitinib 30 mg once daily dose is not recommended for patients with atopic dermatitis receiving chronic treatment with strong CYP3A4 inhibitors. For patients with ulcerative colitis or Crohn's disease using strong CYP3A4 inhibitors, the recommended induction dose is 30 mg once daily and the recommended maintenance dose is 15 mg once daily. Alternatives to strong CYP3A4 inhibitor medications should be considered when used in the long-term. Food or drink containing grapefruit should be avoided during treatment with upadacitinib.

Coadministration with CYP3A4 Inducers

Upadacitinib exposure is decreased when co-administered with strong CYP3A4 inducers (such as rifampin and phenytoin), which may lead to reduced therapeutic effect of upadacitinib. In a clinical study, coadministration of upadacitinib after multiple doses of rifampicin (strong CYP3A inducer) resulted in approximately 50% and 60% decreases in upadacitinib C_{max} and AUC, respectively. Patients should be monitored for changes in disease activity if upadacitinib is co-administered with strong CYP3A4 inducers.

Methotrexate and pH modifying medicinal products (e.g., antacids or proton pump inhibitors) have no effect on upadacitinib plasma exposures.

Potential for upadacitinib to affect the pharmacokinetics of other medicinal products

Administration of multiple 30 mg or 45 mg once daily doses of upadacitinib to healthy subjects had a limited effect on midazolam (sensitive substrate for CYP3A) plasma exposures (24- 26% decrease in midazolam AUC and C_{max}), indicating that upadacitinib 30 mg or 45 mg once daily may have a weak induction effect on CYP3A. In a clinical study, rosuvastatin and atorvastatin AUC were decreased by 33% and 23%, respectively, and rosuvastatin C_{max} was decreased by 23% following the administration of multiple 30 mg once daily doses of upadacitinib to healthy subjects. Upadacitinib had no relevant effect on atorvastatin C_{max} or on plasma exposures of ortho-hydroxyatorvastatin (major active metabolite for atorvastatin). Administration of multiple 45 mg once daily doses of upadacitinib to healthy subjects led to a limited increase in AUC and C_{max} of dextromethorphan (sensitive CYP2D6 substrate) by 30% and 35%, respectively, indicating that upadacitinib 45 mg once daily has a weak inhibitory effect on CYP2D6. No dose adjustment is recommended for CYP3A substrates, CYP2D6 or for rosuvastatin or atorvastatin when coadministered with upadacitinib.

Upadacitinib has no relevant effects on plasma exposures of ethinylestradiol, levonorgestrel, methotrexate, or medicinal products that are substrates for metabolism by CYP1A2, CYP2B6, CYP2C9, or CYP2C19.

Fertility, pregnancy and lactation

Women of Childbearing Potential

Women of childbearing potential should be advised to use effective contraception during treatment and for 4 weeks following the final dose of upadacitinib. Female paediatric patients and/or their parents/caregivers should be informed about the need to contact the treating physician once the patient experiences menarche while taking upadacitinib.

Pregnancy

There are no or limited data on the use of upadacitinib in pregnant women. Studies in animals have shown reproductive toxicity (see **Preclinical safety data** section). Upadacitinib was teratogenic in rats and rabbits with effects in bones in rat foetuses and in the heart in rabbit foetuses when exposed *in utero*.

Upadacitinib is contraindicated during pregnancy (see **Contraindications** section).

If a patient becomes pregnant while taking upadacitinib the parents should be informed of the potential risk to the foetus.

Breast-feeding

It is unknown whether upadacitinib/metabolites are excreted in human milk. Available pharmacodynamic/toxicological data in animals have shown excretion of upadacitinib in milk (see **Preclinical safety data** section).

A risk to newborns/infants cannot be excluded.

Upadacitinib should not be used during breast-feeding. A decision must be made whether to discontinue breast-feeding or to discontinue upadacitinib therapy taking into account the benefit of breast-feeding for the child and the benefit of therapy for the woman.

Fertility

The effect of upadacitinib on human fertility has not been evaluated. Animal studies do not indicate effects with respect to fertility (see **Preclinical safety data** section).

Effects on ability to drive and use machines

Upadacitinib has no or negligible influence on the ability to drive and use machines.

Undesirable effects

Summary of the safety profile

In rheumatologic disease clinical trials, the most commonly reported adverse drug reactions (ADRs) with upadacitinib 15 mg were upper respiratory tract infections, blood creatine phosphokinase (CPK) increased, alanine transaminase increased, bronchitis, nausea, neutropaenia, cough, aspartate transaminase increased, and hypercholesterolaemia.

In the placebo-controlled atopic dermatitis clinical trials, the most commonly reported adverse reactions with upadacitinib 15 mg or 30 mg were upper respiratory tract infection, acne, herpes simplex, headache, blood CPK increased, cough, folliculitis, abdominal pain, nausea, neutropaenia, pyrexia, and influenza.

In the placebo-controlled ulcerative colitis and Crohn's disease induction and maintenance clinical trials, the most commonly reported adverse reactions with upadacitinib 45 mg, 30 mg or 15 mg were upper respiratory tract infection, pyrexia, blood CPK increased, anemia, headache, acne, herpes zoster, neutropaenia, rash, pneumonia, hypercholesterolemia, bronchitis, aspartate transaminase increased, fatigue, folliculitis, alanine transaminase increased, herpes simplex, and influenza.

The most common serious adverse reactions were serious infections (see **Special warnings and precautions for use** section).

The safety profile of upadacitinib with long-term treatment was generally similar to the safety profile during the placebo-controlled period across indications.

Tabulated list of adverse reactions

The following list of adverse reactions is based on experience from clinical studies. The frequency of adverse reactions listed below is defined using the following convention: very common (≥ 1/10); common (≥ 1/100 to < 1/10); uncommon (≥ 1/1,000 to < 1/100). The frequencies in Table 3 are based on the higher of the rates for adverse reactions reported with RINVOQ® in clinical trials of rheumatologic disease (15 mg), atopic dermatitis (15 mg and 30 mg), ulcerative colitis (15 mg, 30 mg and 45 mg), or Crohn's disease (15 mg, 30 mg, and 45 mg). When notable differences in frequency were observed between indications, these are presented in the footnotes below the table.

Table 3. Adverse reactions

System Organ Class	Very common	Common	Uncommon
Infections and infestations	Upper respiratory tract infections (URTI)[a]	Bronchitis[a,b] Herpes zoster[a] Herpes simplex[a] Folliculitis Influenza Urinary tract infection Pneumonia[a,h]	Oral candidiasis Diverticulitis Sepsis
Neoplasms benign, malignant and unspecified (including cysts and polyps)		Non-melanoma skin cancer[f]	
Blood and lymphatic system disorders		Anaemia[a] Neutropaenia[a] Lymphopaenia	
Immune system disorders		Urticaria[c,g]	Serious hypersensitivity reactions[a,e]
Metabolism and nutrition disorders		Hypercholesterolaemia[a,b] Hyperlipidaemia[a,b]	Hypertriglyceridaemia

Respiratory, thoracic and mediastinal disorders		Cough	
Gastrointestinal disorders		Abdominal pain[a,d] Nausea	Gastrointestinal perforation[i]
Skin and subcutaneous tissue disorders	Acne[a,c,d,g]	Rash[a]	
General disorders and administration site conditions		Fatigue Pyrexia Peripheral oedema[a,k]	
Investigations		Blood CPK increased ALT increased[b] AST increased[b] Weight increased[g]	
Nervous system disorders		Headache[a,j]	

[a] Presented as grouped term
[b] In atopic dermatitis trials, the frequency of bronchitis, hypercholesterolaemia, hyperlipidaemia, ALT increased, and AST increased was uncommon.
[c] In rheumatologic disease trials, the frequency was common for acne and uncommon for urticaria.
[d] In ulcerative colitis trials, the frequency was common for acne; abdominal pain was less frequent for upadacitinib than for placebo.
[e] Serious hypersensitivity reactions including anaphylactic reaction and angioedema
[f] Most events reported as basal cell carcinoma and squamous cell carcinoma of skin
[g] In Crohn's disease, the frequency was common for acne, and uncommon for urticaria and weight increased.
[h] Pneumonia was common in Crohn's disease and uncommon across other indications.
[i] Frequency is based on Crohn's disease clinical trials.
[j] Headache was very common in the giant cell arteritis trial.
[k] Frequency is based on the giant cell arteritis trial.

Description of selected adverse reactions
Rheumatoid arthritis

Infections

In placebo-controlled clinical studies with background DMARDs, the frequency of infection over 12/14 weeks in the upadacitinib 15 mg group was 27.4% compared to 20.9% in the placebo group. In methotrexate (MTX)-controlled studies, the frequency of infection over 12/14 weeks in the upadacitinib 15 mg monotherapy group was 19.5% compared to 24.0% in the MTX group. The overall long-term rate of infections for the upadacitinib 15 mg group across all five Phase 3 clinical studies (2,630 patients) was 93.7 events per 100 patient-years.

In placebo-controlled clinical studies with background DMARDs, the frequency of serious infection over 12/14 weeks in the upadacitinib 15 mg group was 1.2% compared to 0.6% in the placebo group. In MTX-controlled studies, the frequency of serious infection over 12/14 weeks in the upadacitinib 15 mg monotherapy group was 0.6% compared to 0.4% in the MTX group. The overall long-term rate of serious infections for the upadacitinib 15 mg group across all five Phase 3 clinical studies was 3.8 events per 100 patient-years. The most common serious infection was pneumonia. The rate of serious infections remained stable with long-term exposure.

There was a higher rate of serious infections in patients ≥ 75 years of age, although data are limited.

Opportunistic infections (excluding tuberculosis)

In placebo-controlled clinical studies with background DMARDs, the frequency of opportunistic infections over 12/14 weeks in the upadacitinib 15 mg group was 0.5% compared to 0.3% in the placebo group. In MTX-controlled studies, there were no cases of opportunistic infection over 12/14 weeks in the upadacitinib 15 mg monotherapy group and 0.2% in the MTX group. The overall long-term rate of opportunistic infections for the upadacitinib 15 mg group across all five Phase 3 clinical studies was 0.6 events per 100 patient-years.

The long-term rate of herpes zoster for the upadacitinib 15 mg group across all five Phase 3 clinical studies was 3.7 events per 100 patient-years. Most of the herpes zoster events involved a single dermatome and were non-serious.

Hepatic transaminase elevations

In placebo-controlled studies with background DMARDs, for up to 12/14 weeks, alanine transaminase (ALT) and aspartate transaminase (AST) elevations ≥ 3 x upper limit of normal (ULN) in at least one measurement were observed in 2.1% and 1.5% of patients treated with upadacitinib 15 mg, compared to 1.5% and 0.7%, respectively, of patients treated with placebo. Most cases of hepatic transaminase elevations were asymptomatic and transient.

In MTX-controlled studies, for up to 12/14 weeks, ALT and AST elevations ≥ 3 x ULN in at least one measurement were observed in 0.8% and 0.4% of patients treated with upadacitinib 15 mg, compared to 1.9% and 0.9%, respectively, of patients treated with MTX.

The pattern and incidence of elevation in ALT/AST remained stable over time including in long term extension studies.

Lipid elevations

Upadacitinib 15 mg treatment was associated with increases in lipid parameters including total cholesterol, triglycerides, LDL cholesterol and HDL cholesterol. There was no change in the LDL/HDL ratio. Elevations were observed at 2 to 4 weeks of treatment and remained stable with longer-term treatment. Among patients in the controlled studies with baseline values below the specified limits, the following frequencies of patients were observed to shift to above the specified limits on at least one occasion during 12/14 weeks (including patients who had an isolated elevated value):

- Total cholesterol ≥ 5.17 mmol/L (200 mg/dL): 62% vs. 31%, in the upadacitinib 15 mg and placebo groups, respectively
- LDL cholesterol ≥ 3.36 mmol/L (130 mg/dL): 42% vs. 19%, in the upadacitinib 15 mg and placebo groups, respectively
- HDL cholesterol ≥ 1.03 mmol/L (40 mg/dL): 89% vs. 61%, in the upadacitinib 15 mg and placebo groups, respectively
- Triglycerides ≥ 2.26 mmol/L (200 mg/dL): 25% vs. 15%, in the upadacitinib 15 mg and placebo groups, respectively

Creatine phosphokinase

In placebo-controlled studies with background DMARDs, for up to 12/14 weeks, increases in CPK values were observed. CPK elevations > 5 x upper limit of normal (ULN) were reported in 1.0% and 0.3% of patients over 12/14 weeks in the upadacitinib 15 mg and placebo groups, respectively. Most elevations > 5 x ULN were transient and did not require treatment discontinuation. Mean CPK values increased by 4 weeks with a mean increase of 60 U/L at 12 weeks and then remained stable at an increased value thereafter including with extended therapy.

Neutropaenia

In placebo-controlled studies with background DMARDs, for up to 12/14 weeks, decreases in neutrophil counts below 1 x 10^9 cells/L in at least one measurement occurred in 1.1% and <0.1% of patients in the upadacitinib 15 mg and placebo groups, respectively. In clinical studies, treatment was interrupted in response to ANC < 1 x 10^9 cells/L (see **Posology and method of administration** section). Mean neutrophil counts decreased over 4 to 8 weeks. The decreases in neutrophil counts remained stable at a lower value than baseline over time including with extended therapy.

Psoriatic Arthritis
Overall, the safety profile observed in patients with active psoriatic arthritis treated with upadacitinib 15 mg was consistent with the safety profile observed in patients with rheumatoid arthritis. A higher incidence of acne and bronchitis was observed in patients treated with upadacitinib 15 mg (1.3% and 3.9%, respectively) compared to placebo (0.3% and 2.7%, respectively).A higher rate of serious infections (2.6 events per 100 patient-years and 1.3 events per 100 patient-years, respectively) and hepatic transaminase elevations (ALT elevations Grade 3 and higher rates 1.4% and 0.4%, respectively) was observed in patients treated with upadacitinib in combination with MTX therapy compared to patients treated with monotherapy. There was a higher rate of serious infections in patients ≥ 65 years of age, although data are limited.

Non-radiographic Axial Spondyloarthritis
A total of 187 patients with non-radiographic axial spondyloarthritis were treated with upadacitinib 15mg in the clinical study representing 116.6 patient-years of exposure, of whom 35 were exposed to upadacitinib 15 mg for at least one year.

Overall, the safety profile observed in patients with active non-radiographic axial spondyloarthritis treated with upadacitinib 15 mg was consistent with the safety profile observed in patients with rheumatoid arthritis. No new safety findings were identified.

Ankylosing Spondylitis
Overall, the safety profile observed in patients with active ankylosing spondylitis treated with upadacitinib 15 mg was consistent with the safety profile observed in patients with rheumatoid arthritis. No new safety findings were identified.

Giant cell arteritis

Overall, the safety profile observed in patients with giant cell arteritis treated with upadacitinib 15 mg was generally consistent with the known safety profile for upadacitinib.

Serious Infections

In the placebo-controlled clinical study, the frequency of serious infections over 52 weeks was 5.7% in the upadacitinib 15 mg group and 10.7% in the placebo group. The long-term rate of serious infections for the upadacitinib 15 mg group was 2.9 events per 100 patient-years.

Opportunistic infections (excluding tuberculosis)

In the placebo-controlled clinical study, the frequency of opportunistic infection (excluding tuberculosis and herpes zoster) over 52 weeks was 1.9% in the upadacitinib 15 mg group and 0.9% in the placebo group. The long-term rate of opportunistic infections (excluding tuberculosis and herpes zoster) for the upadacitinib 15 mg group was 0.6 events per 100 patient-years.

In the placebo-controlled clinical study, the frequency of herpes zoster over 52 weeks was 5.3% in the upadacitinib 15 mg group and 2.7% in the placebo group. The long-term rate of herpes zoster for the upadacitinib 15 mg group was 4.1 events per 100 patient-years.

Atopic dermatitis

Infections
In the placebo-controlled period of the clinical studies, the frequency of infection over 16 weeks in the upadacitinib 15 mg and 30 mg groups was 39% and 43% compared to 30% in the placebo group, respectively. The long-term rate of infections for the upadacitinib 15 mg and 30 mg groups was 98.5 and 109.6 events per 100 patient-years, respectively.

In placebo-controlled clinical studies, the frequency of serious infection over 16 weeks in the upadacitinib 15 mg and 30 mg groups was 0.8% and 0.4% compared to 0.6% in the placebo group, respectively. The long-term rate of serious infections for the upadacitinib 15 mg and 30 mg groups was 2.3 and 2.8 events per 100 patient-years, respectively.

Opportunistic infections (excluding tuberculosis)

In the placebo-controlled period of the clinical studies, all opportunistic infections (excluding TB and herpes zoster) reported were eczema *herpeticum.* The frequency of eczema *herpeticum* over 16 weeks in the upadacitinib 15 mg and 30 mg groups was 0.7% and 0.8% compared to 0.4% in the placebo group, respectively. The long-term rate of eczema *herpeticum* for the upadacitinib 15 mg and 30 mg groups was 1.6 and 1.8 events per 100 patient-years, respectively. One case of esophageal candidiasis was reported with upadacitinib 30 mg.

The long-term rate of herpes zoster for the upadacitinib 15 mg and 30 mg groups was 3.5 and 5.2 events per 100 patient-years, respectively. Most of the herpes zoster events involved a single dermatome and were non-serious.

Laboratory abnormalities

Dose-dependent changes in ALT increased and/or AST increased (≥ 3 x ULN), lipid parameters, CPK values (> 5 x ULN), and neutropaenia *(*ANC < 1 x 10^9 cells/L) associated with upadacitinib treatment were similar to what was observed in the rheumatologic disease clinical studies.

Small increases in LDL cholesterol were observed after week 16 in atopic dermatitis studies.

Ulcerative colitis

The overall safety profile observed in patients with ulcerative colitis was generally consistent with that observed in patients with rheumatoid arthritis.

A higher rate of herpes zoster was observed with an induction treatment period of 16 weeks vs 8 weeks.

RINVOQ has been studied in patients with moderately to severely active UC in one Phase 2b and three Phase 3 (UC-1, UC-2 and UC-3) randomized, double-blind, placebo-controlled clinical studies and a long-term extension study (*see* **CLINICAL STUDIES**) with a total of 1313 patients representing 3537 patient-years of exposure, of whom a total of 959 patients were exposed for at least one year.

In the induction studies (Phase 2b, UC-1, and UC-2), 719 patients received at least one dose of RINVOQ 45 mg, of whom 513 were exposed for 8 weeks and 127 subjects were exposed for up to 16 weeks.

In the maintenance study UC-3 and the long-term extension study, 285 patients received at least one dose of RINVOQ 15 mg, of whom 193 were exposed for at least one year and 291 patients received at least one dose of RINVOQ 30 mg, of whom 214 were exposed for at least one year.

The frequency of adverse reactions listed below is defined using the following convention: very common (≥ 1/10); common (≥ 1/100 to < 1/10); uncommon (≥ 1/1,000 to < 1/100). Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Table 4. Adverse Drug Reactions

System Organ Class	**Very common**	**Common**	**Uncommon**
Infections and infestations	Upper respiratory tract infections (URTI)[a]	Herpes zoster[a] Herpes simplex[a] Folliculitis Influenza	Pneumonia[a]
Neoplasms benign, malignant and unspecified (including cysts and polyps)		Non-melanoma skin cancer[a]	
Blood and lymphatic system disorders		Neutropaenia[a] Lymphopaenia[a]	
Metabolism and nutrition disorders		Hypercholesterolaemia[a] Hyperlipidaemia[a]	
Skin and subcutaneous tissue disorders		Acne[a] Rash[a]	
General disorders and administration site conditions		Pyrexia	
Investigations		Blood CPK increased ALT increased AST increased	
[a] Presented as grouped term			

The safety profile of RINVOQ with long-term treatment was consistent with that in the placebo-controlled period.

Infections

In the placebo-controlled induction studies, the frequency of infection over 8 weeks in the upadacitinib 45 mg group compared to the placebo group was 20.7% and 17.5%, respectively. In the placebo-controlled maintenance study, the frequency of infection over 52 weeks in the upadacitinib 15 mg and 30 mg groups was 40.4% and 44.2%, respectively, compared to 38.8% in the placebo group. The long-term rate of infections for upadacitinib 15 mg and 30 mg was 64.5 and 77.8 events per 100 patient-years, respectively.

In the placebo-controlled induction studies, the frequency of serious infection over 8 weeks in both the upadacitinib 45 mg group and the placebo group was 1.3%. No additional serious infections were observed over 8-week extended treatment with upadacitinib 45 mg. In the placebo-controlled maintenance study, the frequency of serious infection over 52 weeks in the upadacitinib 15 mg and 30 mg groups was 3.6% and 3.2%, respectively, compared to 3.3% in the placebo group. The long-term rate of serious infections for the upadacitinib 15 mg and 30 mg groups was 3.0 and 4.6 events per 100 patient-years, respectively. The most frequently reported serious infection in the induction and maintenance phases was COVID-19 pneumonia.

Opportunistic infections (excluding tuberculosis)
In the placebo-controlled induction studies over 8 weeks, the frequency of opportunistic infection (excluding tuberculosis and herpes zoster) in the upadacitinib 45 mg group was 0.4% and 0.3% in the placebo group. No additional opportunistic infections (excluding tuberculosis and herpes zoster) were observed over 8-week extended treatment with upadacitinib 45 mg. In the placebo-controlled maintenance study over 52 weeks, the frequency of opportunistic infection (excluding tuberculosis and herpes zoster) in the upadacitinib 15 mg and 30 mg groups was 0.8% on placebo and in the upadacitinib 15 mg and 30 mg groups. The long-term rate of opportunistic infections (excluding tuberculosis and herpes zoster) for the upadacitinib 15 mg and 30 mg groups was 0.3 and 0.6 events per 100 patient-years, respectively.

In the placebo-controlled induction studies over 8 weeks, the frequency of herpes zoster in the upadacitinib 45 mg group was 0.6% and 0% in the placebo group. The frequency of herpes zoster was 3.9% over 16-week treatment with upadacitinib 45 mg. In the placebo-controlled maintenance study over 52 weeks, the frequency of herpes zoster in the upadacitinib 15 mg and 30 mg groups was 4.4% and 4.0%, respectively, compared to 0% in the placebo group. The long-term rate of herpes zoster for the upadacitinib 15 mg and 30 mg groups was 4.5 and 7.2 events per 100 patient-years, respectively.

Malignancy
In the placebo-controlled induction studies, there were no reports of malignancy. In the placebo-controlled maintenance study, the frequency of malignancies excluding NMSC in the RINVOQ 15 mg and 30 mg groups was 0.4% and 0.8%, respectively, and 0.4% in the placebo group. The long-term incidence rate of malignancies excluding NMSC for the RINVOQ 15 mg and 30 mg was 0.7 and 0.4 per 100 patient years, respectively.

Gastrointestinal Perforations
In the placebo-controlled maintenance period, gastrointestinal perforation was reported in 1 patient treated with placebo (1.5 per 100 patient-years) and no patients treated with RINVOQ 15 mg or 30 mg. In the long-term extension study, 1 patient treated with RINVOQ 15 mg (0.1 per 100 patient-years) and 1 patient treated with RINVOQ 30 mg (<0.1 per 100 patient-years) reported such events.

Thrombosis
In the placebo-controlled induction studies, the frequency of venous thrombosis (pulmonary embolism or deep vein thrombosis) over 8 weeks in the RINVOQ 45 mg group was 0.1% and 0.3% in the placebo group, respectively. No additional events of venous thrombosis were reported with RINVOQ 45 mg extended induction treatment. In the placebo-controlled maintenance study, the frequency of venous thrombosis over 52 weeks in the RINVOQ 15 mg and 30 mg groups was 0.8% and 0.8%, respectively, and 0% in the placebo group. The long-term incidence rate of venous thrombosis for RINVOQ 15 mg and 30 mg was 0.7 and 0.6 per 100 patient-years, respectively.

Hepatic transaminase elevations
In the placebo-controlled induction studies over 8 weeks, alanine transaminase (ALT) and aspartate transaminase (AST) elevations ≥ 3 x upper limit of normal (ULN) in at least one measurement were observed in 1.5% and 1.5% of patients treated with RINVOQ 45 mg and 0% and 0.3% with placebo, respectively. In the placebo-controlled maintenance study over 52 weeks, ALT elevations ≥ 3 x ULN in at least one measurement were observed in 2.0% and 4.4% of patients treated with RINVOQ 15 mg and 30 mg and 1.2% with placebo, respectively. AST elevations ≥ 3 x ULN in at least one measurement were observed in 1.6% and 2.0% of patients treated with RINVOQ 15 mg and 30 mg and 0.4% with placebo, respectively. Most cases of hepatic transaminase elevations were asymptomatic and transient. The pattern and incidence of ALT/AST elevations remained generally stable over time including in long-term extension studies.

Lipid elevations
Elevations in lipid parameters were observed at 8 weeks of treatment with RINVOQ 45 mg and remained generally stable with longer-term treatment with RINVOQ 15 mg and 30 mg. Among patients in the placebo-controlled induction studies with baseline values below the specified limits, the

following frequencies of patients were observed to shift to above the specified limits on at least one occasion during 8 weeks (including patients who had an isolated elevated value):

- Total cholesterol ≥ 5.17 mmol/L (200 mg/dL): 49% vs. 11%, in the RINVOQ 45 mg and placebo groups, respectively
- LDL cholesterol ≥ 3.36 mmol/L (130 mg/dL): 27% vs. 9%, in the RINVOQ 45 mg and placebo groups, respectively
- HDL cholesterol ≥ 1.03 mmol/L (40 mg/dL): 79% vs. 36%, in the RINVOQ 45 mg and placebo groups, respectively
- Triglycerides ≥ 2.26 mmol/L (200 mg/dL): 6% vs 4% in the RINVOQ 45 mg and placebo groups, respectively

Creatine phosphokinase elevations
In the placebo-controlled induction studies over 8 weeks, increases in creatine phosphokinase (CPK) values were observed. CPK elevations > 5 x ULN were reported in 2.2% and 0.3% of patients in the RINVOQ 45 mg and placebo groups, respectively. In the placebo-controlled maintenance study over 52 weeks, CPK elevations > 5 x ULN were reported in 4.4% and 6.8% of patients in the RINVOQ 15 mg and 30 mg groups and 1.2% in the placebo group, respectively. Most elevations > 5 x ULN were transient and did not require treatment discontinuation.

Neutropenia
In the placebo-controlled induction studies over 8 weeks, decreases in neutrophil counts below 1000 cells/mm^3 in at least one measurement occurred in 2.8% of patients in the RINVOQ 45 mg group and 0% in the placebo group, respectively. In the placebo-controlled maintenance study over 52 weeks, decreases in neutrophil counts below 1000 cells/mm^3 in at least one measurement occurred in 0.8% and 2.4% of patients in the RINVOQ 15 mg and 30 mg groups and 0.8% in the placebo group, respectively.

Lymphopenia
In the placebo-controlled induction studies over 8 weeks, decreases in lymphocyte counts below 500 cells/mm^3 in at least one measurement occurred in 2.0% of patients in the RINVOQ 45 mg group and 0.8% in the placebo group. In the placebo-controlled maintenance study over 52 weeks, decreases in lymphocyte counts below 500 cells/mm^3 in at least one measurement occurred in 1.6% and 1.2% of patients in the RINVOQ 15 mg and 30 mg groups and to 0.8% in the placebo group, respectively.

Anemia
In the placebo-controlled induction studies over 8 weeks, hemoglobin decreases below 8 g/dL in at least one measurement occurred in 0.3% of patients in the RINVOQ 45 mg group and 2.1% in the placebo group. In the placebo-controlled maintenance study over 52 weeks, hemoglobin decreases below 8 g/dL in at least one measurement occurred in 0% and 0.4% of patients in the RINVOQ 15 mg and 30 mg groups and 1.2% in the placebo group, respectively.

Crohn's disease

Overall, the safety profile observed in patients with Crohn's disease treated with upadacitinib was consistent with the known safety profile for upadacitinib.

Serious infections

In the placebo-controlled induction studies, the frequency of serious infection over 12 weeks in the upadacitinib 45 mg group and the placebo group was 1.9% and 1.7%, respectively. In the placebo-controlled maintenance study, the frequency of serious infection over 52 weeks in the upadacitinib 15 mg and 30 mg groups was 3.2% and 5.7%, respectively, compared to 4.5% in the placebo group. The long-term rate of serious infections for the upadacitinib 15 mg and 30 mg groups in patients who responded to upadacitinib 45 mg as induction treatment was 5.1 and 7.3 events per 100 patient-years, respectively. The most frequently reported serious infection in the induction and maintenance studies was gastrointestinal infections.

Gastrointestinal Perforations

During the placebo-controlled period in the Phase 3 induction clinical studies, gastrointestinal perforation was reported in 1 patient (0.1%) treated with upadacitinib 45 mg and no patients on placebo through 12 weeks. In all patients treated with upadacitinib 45 mg (n=938) during the induction studies, gastrointestinal perforation was reported in 4 patients (0.4%).

In the long-term placebo-controlled period, gastrointestinal perforation was reported in 1 patient each treated with placebo (0.7 per 100 patient-years), upadacitinib 15 mg (0.4 per 100 patient-years), and upadacitinib 30 mg (0.4 per 100 patient-years). In all patients treated with rescue upadacitinib 30 mg (n=336), gastrointestinal perforation was reported in 3 patients (0.8 per 100 patient-years) through long-term treatment.

Laboratory abnormalities

In the induction and maintenance clinical studies, the laboratory changes in ALT increased and/or AST increased (≥ 3 x ULN), CPK values (> 5 x ULN), neutropaenia (ANC < 1 x 10^9 cells/L), and lipid parameters associated with upadacitinib treatment were generally similar to what was observed in the rheumatologic disease, atopic dermatitis and ulcerative colitis clinical studies. Dose-dependent changes for these laboratory parameters associated with 15 mg and 30 mg upadacitinib treatment were observed.

In the placebo-controlled induction studies for up to 12 weeks, decreases in lymphocyte counts below 0.5 x 10^9 cells/L in at least one measurement occurred in 2.2% and 2.0% of patients in the upadacitinib 45 mg and placebo groups, respectively. In the placebo-controlled maintenance study, for up to 52 weeks, decreases in lymphocyte counts below 0.5 x 10^9 cells/L in at least one measurement occurred in 4.6%, 5.2% and 1.8% of patients in the upadacitinib 15 mg, 30 mg and placebo groups, respectively. In clinical studies, treatment was interrupted in response to ALC < 0.5 x 10^9 cells/L (see **posology and method of administration**). No notable mean changes of lymphocyte counts were observed during upadacitinib treatment over time.

In the placebo-controlled induction studies for up to 12 weeks, decreases in haemoglobin concentration to below 8 g/dL in at least one measurement occurred in 2.7% and 1.4% of patients in the upadacitinib 45 mg and placebo groups, respectively. In the placebo-controlled maintenance study, for up to 52 weeks, decreases in haemoglobin concentration below 8 g/dL in at least one measurement occurred in 1.4%, 4.4% and 2.8% of patients in the upadacitinib 15 mg, 30 mg and placebo groups, respectively. In clinical studies, treatment was interrupted in response to Hb < 8 g/dL (see **posology and method of administration**). No notable mean changes of haemoglobin concentration were observed during upadacitinib treatment over time.

Elderly

Based on limited data in atopic dermatitis patients aged 65 years and older, there was a higher rate of overall adverse reactions-with the upadacitinib 30 mg dose compared to the 15 mg dose.

Based on the limited data in ulcerative colitis and Crohn's disease patients aged 65 years and older, there was a higher rate of overall adverse reactions with the upadacitinib 30 mg dose compared to the 15 mg dose with maintenance treatment (see **Special warnings and precautions for use** section).

Paediatric population

A total of 541 adolescents aged 12 to 17 years weighing at least 40 kg with atopic dermatitis were treated in the global Phase 3 studies (n=343) and the supplemental adolescent substudies (n=198), of whom 264 were exposed to 15 mg and 265 were exposed to 30 mg. The safety profile for RINVOQ 15 mg and 30 mg was similar in adolescents and adults. With long-term exposure, the adverse drug reaction of skin papilloma was reported in 3.4% and 6.8% of adolescents with atopic dermatitis in the RINVOQ 15 mg and 30 mg groups, respectively.

Reporting of suspected adverse reactions
Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system.

Overdose

Upadacitinib was administered in clinical studies up to doses equivalent in daily AUC to 60 mg extended-release once daily. Adverse reactions were comparable to those seen at lower doses and no specific toxicities were identified. Approximately 90% of upadacitinib in the systemic circulation is eliminated within 24 hours of dosing (within the range of doses evaluated in clinical studies). In case of an overdose, it is recommended that the patient be monitored for signs and symptoms of adverse reactions. Patients who develop adverse reactions should receive appropriate treatment.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties

Pharmacotherapeutic group: Immunosuppressants, selective immunosuppressants ATC code: L04AA44

Mechanism of Action

Upadacitinib is a selective and reversible Janus Kinase (JAK) inhibitor. JAKs are intracellular enzymes that transmit cytokine or growth factor signals involved in a broad range of cellular processes including inflammatory responses, hematopoiesis and immune surveillance. The JAK family of enzymes contains four members, JAK1, JAK2, JAK3 and TYK2 which work in pairs to phosphorylate and activate signal transducers and activators of transcription (STATs). This phosphorylation, in turn, modulates gene expression and cellular function. JAK1 is important in inflammatory cytokine signals while JAK2 is important for red blood cell maturation and JAK3 signals play a role in immune surveillance and lymphocyte function.

In human cellular assays, upadacitinib preferentially inhibits signalling by JAK1 or JAK1/3 with functional selectivity over cytokine receptors that signal via pairs of JAK2. Atopic dermatitis is driven by pro-inflammatory cytokines (including IL-4, IL-13, IL-22, TSLP, IL-31 and IFN-γ) that transduce signals via the JAK1 pathway. Inhibiting JAK1 with upadacitinib reduces the signaling of many mediators which drive the signs and symptoms of atopic dermatitis such as eczematous skin lesions and pruritus.
Pro-inflammatory cytokines (primarily IL-6, IL-7, IL-15 and IFNγ) transduce signals via the JAK1 pathway and are involved in the pathology of inflammatory bowel diseases. JAK1 inhibition with upadacitinib modulates the signaling of the JAK-dependent cytokines underlying the inflammatory burden and signs and symptoms of inflammatory bowel diseases.

Pharmacodynamic effects

Inhibition of IL-6 Induced STAT3 and IL-7 Induced STAT5 Phosphorylation

In healthy volunteers, the administration of upadacitinib (immediate release formulation) resulted in a dose- and concentration-dependent inhibition of IL-6 (JAK1/JAK2) - induced STAT3 and IL-7 (JAK1/JAK3)-induced STAT5 phosphorylation in whole blood. The maximal inhibition was observed 1 hour after dosing which returned to near baseline by the end of dosing interval.

Lymphocytes

In patients with rheumatoid arthritis, treatment with upadacitinib was associated with a small, transient increase in mean ALC from baseline up to Week 36 which gradually returned to at or near baseline levels with continued treatment.

hsCRP

In patients with rheumatoid arthritis, treatment with upadacitinib was associated with decreases from baseline in mean hsCRP levels as early as Week 1 which were maintained with continued treatment.

Vaccine Study

The influence of upadacitinib on the humoral response following administration of adjuvanted recombinant glycoprotein E herpes zoster vaccine was evaluated in 93 patients with rheumatoid arthritis under stable treatment with upadacitinib 15 mg. 98% of patients (n=91) were on concomitant methotrexate. 49% of patients were on oral corticosteroids at baseline. Vaccination resulted in a satisfactory humoral response, 4 weeks post vaccination dose 2, in 88% (95% CI: 81.0, 94.5) of patients treated with upadacitinib 15 mg.

The influence of upadacitinib on the humoral response following the administration of inactivated pneumococcal polysaccharide conjugate vaccine (13-valent, adsorbed) was evaluated in 111 patients with rheumatoid arthritis under stable treatment with upadacitinib 15 mg (n=87) or 30 mg (n=24). 97% of patients (n=108) were on concomitant methotrexate. The primary endpoint was the proportion of patients with satisfactory humoral response defined as ≥ 2-fold increase in antibody concentration from baseline to Week 4 in at least 6 out of the 12 pneumococcal antigens (1, 3, 4, 5, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23F). Results at Week 4 demonstrated a satisfactory humoral response in 67.5% (95% CI: 57.4, 77.5) and 56.5% (95% CI: 36.3, 76.8) of patients treated with upadacitinib 15 mg and 30 mg, respectively.

<u>Clinical efficacy and safety</u>

Rheumatoid arthritis

The efficacy and safety of upadacitinib 15 mg once daily was assessed in five Phase 3 randomised, double-blind, multicentre studies in patients with moderately to severely active rheumatoid arthritis and fulfilling the ACR/EULAR 2010 classification criteria (see Table 4). Patients 18 years of age and older were eligible to participate. The presence of at least 6 tender and 6 swollen joints and evidence of systemic inflammation based on elevation of hsCRP was required at baseline. Four studies included long-term extensions for up to 5 years, and one study (SELECT-COMPARE) included a long-term extension for up to 10 years.

The primary analysis for each of these studies included all randomised subjects who received at least 1 dose of upadacitinib or placebo, and non-responder imputation was used for categorical endpoints.

Across the Phase 3 studies, the efficacy seen with upadacitinib 15 mg QD was generally similar to that observed with upadacitinib 30 mg QD.

Table 5: Clinical trials summary

Study name	Population (n)	Treatment arms	Key outcome measures
SELECT-EARLY	MTX-naive[a] (947)	• Upadacitinib 15 mg • Upadacitinib 30 mg • MTX Monotherapy	• Primary endpoint: clinical remission (DAS28-CRP) at week 24 • Low disease activity (DAS28-CRP) • ACR50 • Radiographic progression (mTSS)

			• Physical function (HAQ-DI) • SF-36 PCS
SELECT-MONOTHERAPY	MTX-IR[b] (648)	• Upadacitinib 15 mg • Upadacitinib 30 mg • MTX Monotherapy	• Primary endpoint: low disease activity (DAS28-CRP) at week 14 • Clinical remission (DAS28-CRP) • ACR20 • Physical function (HAQ-DI) • SF-36 PCS • Morning stiffness
SELECT-NEXT	csDMARD-IR[c] (661)	• Upadacitinib 15 mg • Upadacitinib 30 mg • Placebo On background csDMARDs	• Primary endpoint: low disease activity (DAS28-CRP) at week 12 • Clinical remission (DAS28-CRP) • ACR20 • Physical function (HAQ-DI) • SF-36 PCS • Low disease activity (CDAI) • Morning stiffness • FACIT-F
SELECT-COMPARE	MTX-IR[d] (1,629)	• Upadacitinib 15 mg • Placebo • Adalimumab 40 mg On background MTX	• Primary endpoint: clinical remission (DAS28-CRP) at week 12 • Low disease activity (DAS28-CRP) • ACR20 • Low disease activity (DAS28-CRP) vs adalimumab • Radiographic progression (mTSS) • Physical function (HAQ-DI) • SF-36 PCS • Low disease activity (CDAI) • Morning stiffness • FACIT-F
SELECT-BEYOND	bDMARD-IR[e] (499)	• Upadacitinib 15 mg • Upadacitinib 30 mg • Placebo On background csDMARDs	• Primary endpoint: low disease activity (DAS28-CRP) at week 12 • ACR20 • Physical function (HAQ-DI) • SF-36 PCS

Abbreviations: ACR20 (or 50) = American College of Rheumatology ≥20% (or ≥50%) improvement; bDMARD = biologic disease-modifying anti-rheumatic drug, CRP = C-Reactive Protein, DAS28 = Disease Activity Score 28 joints, mTSS = modified Total Sharp Score, csDMARD = conventional synthetic disease-modifying anti-rheumatic drug, HAQ-DI = Health Assessment Questionnaire Disability Index, SF-36 PCS = Short Form (36) Health Survey (SF-36) Physical Component Summary, CDAI = Clinical Disease Activity Index, FACIT-F = Functional Assessment of Chronic Illness Therapy-Fatigue score, IR = inadequate responder, MTX = methotrexate, n = number randomised

[a.] Patients were naïve to MTX or received no more than 3 weekly MTX doses

[b] Patients had inadequate response to MTX

[c] Patients who had an inadequate response to csDMARDs; patients with prior exposure to at most one bDMARD were eligible (up to 20% of total number of patients) if they had either limited exposure (<3 months) or had to discontinue the bDMARD due to intolerability

[d] Patients who had an inadequate response to MTX; patients with prior exposure to at most one bDMARD (except adalimumab) were eligible (up to 20% of total study number of patients) if they had either limited exposure (<3 months) or had to discontinue the bDMARD due to intolerability

[e] Patients who had an inadequate response or intolerance to at least one bDMARD

Clinical Response:

Remission and low disease activity

In the studies, a significantly higher proportion of patients treated with upadacitinib 15 mg achieved low disease activity (DAS28-CRP ≤3.2) and clinical remission (DAS28-CRP <2.6) compared to placebo, MTX or adalimumab (Table 5). Compared to adalimumab, significantly higher rates of low disease activity were achieved at week 12 in SELECT-COMPARE. Overall, both low disease activity and clinical remission rates were consistent across patient populations, with or without MTX. At 3 years, 297/651 (45.6%) and 111/327 (33.9%) patients remained on originally randomised treatment of upadacitinib 15 mg or adalimumab, respectively, in SELECT-COMPARE, and 216/317 (68.1%) and 149/315 (47.3%) patients remained on originally randomised treatment of upadacitinib 15 mg or MTX monotherapy, respectively, in SELECT-EARLY. Among the patients who remained on their originally allocated treatment, low disease activity and clinical remission were maintained through 3 years.

ACR Response

In all studies, more patients treated with upadacitinib 15 mg achieved ACR20, ACR50, and ACR70 responses at 12 weeks compared to placebo, MTX, or adalimumab (Table 6). Time to onset of efficacy was rapid across measures with greater responses seen as early as week 1 for ACR20. Durable response rates were observed (with or without MTX), with ACR20/50/70 responses maintained through 3 years among the patients who remained on their originally allocated treatment.

Treatment with upadacitinib 15 mg, alone or in combination with csDMARDs, resulted in improvements in individual ACR components, including tender and swollen joint counts, patient and physician global assessments, HAQ-DI, pain assessment and hsCRP.

Table 6: Response and remission

Study	**SELECT EARLY MTX-Naïve**		**SELECT MONO MTX-IR**		**SELECT NEXT csDMARD-IR**		**SELECT COMPARE MTX-IR**			**SELECT BEYOND bDMARD-IR**	
	MTX	UPA 15mg	MTX	UPA 15mg	PBO	UPA 15mg	PBO	UPA 15mg	ADA 40mg	PBO	UPA 15mg
N	314	317	216	217	221	221	651	651	327	169	164
Week											
LDA DAS28-CRP ≤3.2 (% of patients)											
12[a]/14[b]	28	53[g]	19	45[e]	17	48[e]	14	45[e,h]	29	14	43[e]
24[c]26[d]	32	60[f]					18	55[g,h]	39		
48	39	59[g]						50[h]	35		
CR DAS28-CRP <2.6 (% of patients)											
12[a]/14[b]	14	36[g]	8	28[e]	10	31[e]	6	29[e,h]	18	9	29[g]
24[c]26[d]	18	48[e]					9	41[g,h]	27		
48	29	49[g]						38[i]	28		
ACR20 (% of patients)											
12[a]/14[b]	54	76[g]	41	68[e]	36	64[e]	36	71[e,j]	63	28	65[e]
24[c]/26[d]	59	79[g]					36	67[g,i]	57		
48	57	74[g]						65[i]	54		
ACR50 (% of patients)											
12[a]/14[b]	28	52[g]	15	42[g]	15	38[g]	15	45[g,h]	29	12	34[g]

24[c]/26[d]	33	60[e]					21	54[g,h]	42		
48	43	63[g]						49[i]	40		
ACR70 (% of patients)											
12[a]/14[b]	14	32[g]	3	23[g]	6	21[g]	5	25[g,h]	13	7	12
24[c]/26[d]	18	44[g]					10	35[g,h]	23		
48	29	51[g]						36[h]	23		
CDAI ≤10 (% of patients)											
12[a]/14[b]	30	46[g]	25	35[l]	19	40[e]	16	40[e,h]	30	14	32[g]
24[c]/26[d]	38	56[g]					22	53[g,h]	38		
48	43	60[g]						47[h]	34		

Abbreviations: ACR20 (or 50 or 70) = American College of Rheumatology ≥20% (or ≥50% or ≥70%) improvement; ADA = adalimumab; bDMARD = biologic disease-modifying anti-rheumatic drug; CDAI = Clinical Disease Activity Index; CR = Clinical Remission; CRP = C-Reactive Protein, csDMARD = conventional synthetic disease-modifying anti-rheumatic drug; DAS28 = Disease Activity Score 28 joints; IR = inadequate responder; LDA = Low Disease Activity; MTX = methotrexate; PBO = placebo; UPA= upadacitinib

[a] SELECT-NEXT, SELECT-EARLY, SELECT-COMPARE, SELECT-BEYOND
[b] SELECT-MONOTHERAPY
[c] SELECT-EARLY
[d] SELECT-COMPARE
[e] multiplicity-controlled $p \leq 0.001$upadacitinib vs placebo or MTX comparison
[f] multiplicity-controlled $p \leq 0.01$ upadacitinib vs placebo or MTX comparison
[g] nominal $p \leq 0.001$ upadacitinib vs placebo or MTX comparison
[h] nominal $p \leq 0.001$ upadacitinib vs adalimumab comparison
[i] nominal $p \leq 0.01$ upadacitinib vs adalimumab comparison
[j] nominal $p < 0.05$ upadacitinib vs adalimumab comparison
[k] nominal $p \leq 0.01$ upadacitinib vs placebo or MTX comparison
[l] nominal $p < 0.05$ upadacitinib vs MTX comparison
Note: Week 48-data derived from analysis on Full Analysis set (FAS) by randomised group using Non-Responder Imputation

Radiographic Response

Inhibition of progression of structural joint damage was assessed using the modified Total Sharp Score (mTSS) and its components, the erosion score and joint space narrowing score, at weeks 24/26 and week 48 in SELECT-EARLY and SELECT-COMPARE.

Treatment with upadacitinib 15 mg resulted in significantly greater inhibition of the progression of structural joint damage compared to placebo in combination with MTX in SELECT-COMPARE and as monotherapy compared to MTX in SELECT-EARLY (Table 7). Analyses of erosion and joint space narrowing scores were consistent with the overall scores. The proportion of patients with no radiographic progression (mTSS change ≤ 0) was significantly higher with upadacitinib 15 mg in both studies. Inhibition of progression of structural joint damage was maintained through Week 96 in both studies for patients who remained on their originally allocated treatment with upadacitinib 15 mg (based on available results from 327 patients in SELECT-COMPARE and 238 patients in SELECT-EARLY).

Table 7: Radiographic changes

Study	SELECT EARLY MTX-Naïve		SELECT COMPARE MTX-IR		
Treatment Group	MTX	UPA 15 mg	PBO[a]	UPA 15 mg	ADA 40 mg
Modified Total Sharps Score, mean change from baseline					
Week 24[b]/26[c]	0.7	0.1[f]	0.9	0.2[g]	0.1
Week 48	1.0	0.03[e]	1.7	0.3[e]	0.4
Proportion of patients with no radiographic progression[d]					
Week 24[b]/26[c]	77.7	87.5[f]	76.0	83.5[f]	86.8
Week 48	74.3	89.9[e]	74.1	86.4[e]	87.9

Abbreviations: ADA = adalimumab; IR = inadequate responder; MTX = methotrexate; PBO = placebo; UPA= upadacitinib

[a] All placebo data at week 48 derived using linear extrapolation

[b] SELECT-EARLY

[c] SELECT-COMPARE

[d] No progression defined as mTSS change ≤ 0

[e] nominal p≤0.001 upadacitinib vs placebo or MTX comparison

[f] multiplicity-controlled p≤0.01 upadacitinib vs placebo or MTX comparison

[g] multiplicity-controlled p≤0.001 upadacitinib vs placebo or MTX comparison

Physical Function Response and Health-Related Outcomes

Treatment with upadacitinib 15 mg, alone or in combination with csDMARDs, resulted in a significantly greater improvement in physical function compared to all comparators as measured by HAQ-DI (see Table 8).

Improvements in HAQ-DI and pain were maintained through 3 years for patients who remained on their originally allocated treatment with upadacitinib 15 mg based on available results from SELECT-COMPARE and SELECT-EARLY.

Table 8 Mean change from baseline in HAQ-DI[a,b]

Study	SELECT EARLY MTX-Naïve		SELECT MONO MTX-IR		SELECT NEXT csDMARD-IR		SELECT COMPARE MTX-IR			SELECT BEYOND BIO-IR	
Treatment group	MTX	UPA 15mg	MTX	UPA 15mg	PBO	UPA 15mg	PBO	UPA 15mg	ADA 40mg	PBO	UPA 15mg
N	313	317	216	216	220	216	648	644	324	165	163
Baseline score, mean	1.6	1.6	1.5	1.5	1.4	1.5	1.6	1.6	1.6	1.6	1.7
Week 12[c]/14[d]	-0.5	-0.8[h]	-0.3	-0.7[g]	-0.3	-0.6[g]	-0.3	-0.6[g,i]	-0.5	-0.2	-0.4[g]
Week 24[e]/26[f]	-0.6	-0.9[g]					-0.3	-0.7[h,i]	-0.6		

Abbreviations: ADA = adalimumab; HAQ-DI = Health Assessment Questionnaire Disability Index; IR = inadequate responder; MTX = methotrexate; PBO = placebo; UPA = upadacitinib
[a] Data shown are mean
[b] Health Assessment Questionnaire-Disability Index: 0=best, 3=worst; 20 questions; 8 categories: dressing and grooming, arising, eating, walking, hygiene, reach, grip, and activities.
[c] SELECT-EARLY, SELECT-NEXT, SELECT-COMPARE, SELECT-BEYOND
[d] SELECT-MONOTHERAPY
[e] SELECT-EARLY
[f] SELECT-COMPARE
[g] multiplicity-controlled p≤0.001 upadacitinib vs placebo or MTX comparison
[h] nominal p≤0.001 upadacitinib vs placebo or MTX comparison
[i] nominal p≤0.01 upadacitinib vs adalimumab comparison

In the studies SELECT-MONOTHERAPY, SELECT-NEXT, and SELECT-COMPARE, treatment with upadacitinib 15 mg resulted in a significantly greater improvement in the mean duration of morning joint stiffness compared to placebo or MTX.

In the clinical studies, upadacitinib treated patients reported significant improvements in patient-reported quality of life, as measured by the Short Form (36) Health Survey (SF-36) Physical Component Score compared to placebo and MTX. Moreover, upadacitinib treated patients reported significant improvements in fatigue, as measured by the Functional Assessment of Chronic Illness Therapy-Fatigue score (FACIT-F) compared to placebo.

Psoriatic arthritis

The efficacy and safety of upadacitinib 15 mg once daily were assessed in two Phase 3 randomised, double-blind, multicenter, placebo-controlled studies in patients 18 years of age or older with moderately to severely active psoriatic arthritis. All patients had active psoriatic arthritis for at least 6 months based upon the Classification Criteria for Psoriatic Arthritis (CASPAR), at least 3 tender joints and at least 3 swollen joints, and active plaque psoriasis or history of plaque psoriasis. For both studies, the primary endpoint was the proportion of patients who achieved an ACR20 response at week 12.

SELECT-PsA 1 was a 24-week trial in 1705 patients who had an inadequate response or intolerance to at least one non-biologic DMARD. At baseline, 1393 (82%) of patients were on at least one concomitant non-biologic DMARD; 1084 (64%) of patients received concomitant MTX only; and 311 (18%) of patients were on monotherapy. Patients received upadacitinib 15 mg or 30 mg once daily, adalimumab, or placebo. At week 24, all patients randomised to placebo were switched to upadacitinib 15 mg or 30 mg once daily in a blinded manner. SELECT-PsA 1 included a long-term extension for up to 5 years.

SELECT-PsA 2 was a 24-week trial in 642 patients who had an inadequate response or intolerance to at least one biologic DMARD. At baseline, 296 (46%) of patients were on at least one concomitant non-biologic DMARD; 222 (35%) of patients received concomitant MTX only; and 345 (54%) of patients were on monotherapy. Patients received upadacitinib 15 mg or 30 mg once daily or placebo. At week 24, all patients randomised to placebo were switched to upadacitinib 15 mg or 30 mg once daily in a blinded manner. SELECT-PsA 2 included a long-term extension for up to 3 years.

Clinical response

In both studies, a statistically significant greater proportion of patients treated with upadacitinib 15 mg achieved ACR20 response compared to placebo at week 12 (Table 9). Time to onset of efficacy was rapid across measures with greater responses seen as early as week 2 for ACR20.

Treatment with upadacitinib 15 mg resulted in improvements in individual ACR components, including tender/painful and swollen joint counts, patient and physician global assessments, HAQ-DI, pain assessment, and hsCRP compared to placebo.

In SELECT-PsA 1, upadacitinib 15 mg achieved non-inferiority compared to adalimumab in the proportion of patients achieving ACR20 response at week 12; however, superiority to adalimumab could not be demonstrated.

In both studies, consistent responses were observed alone or in combination with methotrexate for primary and key secondary endpoints.

The efficacy of upadacitinib 15 mg was demonstrated regardless of subgroups evaluated including baseline BMI, baseline hsCRP, and number of prior non-biologic DMARDs (≤ 1 or >1).

Table 9: Clinical response in SELECT-PsA 1 and SELECT-PsA 2

Study	**SELECT-PsA 1 non-biologic DMARD-IR**			**SELECT-PsA 2 bDMARD-IR**	
Treatment Group	PBO	UPA 15 mg	ADA 40 mg	PBO	UPA 15 mg
N	423	429	429	212	211
ACR20, % of patients (95% CI)					
Week 12	36 (32, 41)	71 (66, 75)[f]	65 (61, 70)	24 (18, 30)	57 (50, 64)
Difference from placebo (95% CI)	35 (28, 41)[d,e]		-	33 (24, 42)[d,e]	
Week 24	45 (40, 50)	73 (69, 78)	67 (63, 72)	20 (15, 26)	59 (53, 66)
Week 56		74 (70, 79)	69 (64, 73)		60 (53, 66)
ACR50, % of patients (95% CI)					
Week 12	13 (10, 17)	38 (33, 42)	38 (33, 42)	5 (2, 8)	32 (26, 38)
Week 24	19 (15, 23)	52 (48, 57)	44 (40, 49)	9 (6, 13)	38 (32, 45)
Week 56		60 (55, 64)	51 (47, 56)		41 (34, 47)
ACR70, % of patients (95% CI)					
Week 12	2 (1, 4)	16 (12, 19)	14 (11, 17)	1 (0, 1)	9 (5, 12)
Week 24	5 (3, 7)	29 (24, 33)	23 (19, 27)	1 (0, 2)	19 (14, 25)
Week 56		41 (36, 45)	31 (27, 36)		24 (18, 30)
MDA, % of patients (95% CI)					
Week 12	6 (4, 9)	25 (21, 29)	25 (21, 29)	4 (2, 7)	17 (12, 22)
Week 24	12 (9, 15)	37 (32, 41)[e]	33 (29, 38)	3 (1, 5)	25 (19, 31)[e]
Week 56		45 (40, 50)	40 (35, 44)		29 (23, 36)
Resolution of enthesitis (LEI=0), % of patients (95% CI)[a]					

Week 12	33 (27, 39)	47 (42, 53)	47 (41, 53)	20 (14, 27)	39 (31, 47)
Week 24	32 (27, 39)	54 (48, 60)[e]	47 (42, 53)	15 (9, 21)	43 (34, 51)
Week 56		59 (53, 65)	54 (48, 60)		43 (34, 51)
Resolution of dactylitis (LDI=0), % of patients (95% CI)[b]					
Week 12	42 (33, 51)	74 (66, 81)	72 (64, 80)	36 (24, 48)	64 (51, 76)
Week 24	40 (31, 48)	77 (69, 84)	74 (66, 82)	28 (17, 39)	58 (45, 71)
Week 56		75 (68, 82)	74 (66, 82)		51 (38, 64)
PASI75, % of patients (95% CI)[c]					
Week 16	21 (16, 27)	63 (56, 69)[e]	53 (46, 60)	16 (10, 22)	52 (44, 61)[e]
Week 24	27 (21, 33)	64 (58, 70)	59 (52, 65)	19 (12, 26)	54 (45, 62)
Week 56		65 (59, 72)	61 (55, 68)		52 (44, 61)
PASI90, % of patients (95% CI)[c]					
Week 16	12 (8, 17)	38 (32, 45)	39 (32, 45)	8 (4, 13)	35 (26, 43)
Week 24	17 (12, 22)	42 (35, 48)	45 (38, 52)	7 (3, 11)	36 (28, 44)
Week 56		49 (42, 56)	47 (40, 54)		41 (32, 49)

Abbreviations: ACR20 (or 50 or 70) = American College of Rheumatology ≥20% (or ≥50% or ≥70%) improvement, ADA = adalimumab; bDMARD = biologic disease-modifying anti-rheumatic drug; IR = inadequate responder; MDA = minimal disease activity; PASI75 (or 90) = ≥75% (or ≥90%) improvement in Psoriasis Area and Severity Index; PBO = placebo; UPA= upadacitinib

Patients who discontinued randomised treatment or were missing data at week of evaluation were imputed as non-responders in the analyses. For MDA, resolution of enthesitis, and resolution of dactylitis at week 24/56, the subjects rescued at week 16 were imputed as non-responders in the analyses.

[a] In patients with enthesitis at baseline (n=241, 270, and 265, respectively, for SELECT-PsA 1 and n=144 and 133, respectively, for SELECT-PsA 2)

[b] In patients with dactylitis at baseline (n=126, 136, and 127, respectively, for SELECT-PsA 1 and n=64 and 55, respectively, for SELECT-PsA 2)

[c] In patients with ≥ 3% BSA psoriasis at baseline (n=211, 214, and 211, respectively, for SELECT-PsA 1 and n=131 and 130, respectively, for SELECT-PsA 2)

[d] primary endpoint

[e] multiplicity-controlled p≤0.001 upadacitinib vs placebo comparison

[f] multiplicity-controlled p≤0.001 upadacitinib vs adalimumab comparison (non-inferiority test)

Radiographic response

In SELECT-PsA 1, inhibition of progression of structural damage was assessed radiographically and expressed as the change from baseline in modified Total Sharp Score (mTSS) and its components, the erosion score and the joint space narrowing score, at week 24.

Treatment with upadacitinib 15 mg resulted in statistically significant greater inhibition of the progression of structural joint damage compared to placebo at week 24 (Table 10). Erosion and joint space narrowing scores were consistent with the overall scores. The proportion of patients with no radiographic progression (mTSS change ≤ 0.5) was higher with upadacitinib 15 mg compared to placebo at week 24.

Table 10: Radiographic changes in SELECT-PsA 1

Treatment Group	PBO	UPA 15 mg	ADA 40 mg
Modified Total Sharp Score, mean change from baseline (95% CI)			
Week 24	0.25 (0.13, 0.36)	-0.04 (-0.16, 0.07)[c]	0.01 (-0.11, 0.13)
Week 56[a]	0.44 (0.29, 0.59)	-0.05 (-0.20, 0.09)	-0.06 (-0.20, 0.09)
Proportion of patients with no radiographic progression[b], % (95% CI)			
Week 24	92 (89, 95)	96 (94, 98)	95 (93, 97)
Week 56[a]	89 (86, 92)	97 (96, 99)	94 (92, 97)

Abbreviations: ADA = adalimumab; PBO = placebo; UPA= upadacitinib
[a] All placebo data at week 56 derived using linear extrapolation
[b] No progression defined as mTSS change ≤0.5
[c] multiplicity-controlled p≤0.001 upadacitinib vs placebo comparison

Physical function response and health-related outcomes

In SELECT-PsA 1, patients treated with upadacitinib 15 mg showed statistically significant improvement from baseline in physical function as assessed by HAQ-DI at week 12 (-0.42 [95% CI: -0.47, -0.37]) compared to placebo (-0.14 [95% CI: -0.18, -0.09]); improvement in patients treated with adalimumab was -0.34 (95% CI: -0.38, -0.29). In SELECT-PsA 2, patients treated with upadacitinib 15 mg showed statistically significant improvement from baseline in HAQ-DI at week 12 (-0.30 [95% CI: -0.37, -0.24]) compared to placebo (-0.10 [95% CI: -0.16, -0.03]). Improvement in physical function was maintained through week 56 in both studies.

Health-related quality of life was assessed by SF-36v2. In both studies, patients receiving upadacitinib 15 mg experienced statistically significant greater improvement from baseline in the Physical Component Summary score compared to placebo at week 12. Improvements from baseline were maintained through week 56 in both studies.

Patients receiving upadacitinib 15 mg experienced statistically significant improvement from baseline in fatigue, as measured by FACIT-F score, at week 12 compared to placebo in both studies. Improvements from baseline were maintained through week 56 in both studies.

At baseline, psoriatic spondylitis was reported in 31% and 34% of patients in SELECT-PsA 1 and SELECT-PsA 2, respectively. Patients with psoriatic spondylitis treated with upadacitinib 15 mg showed improvements from baseline in Bath Ankylosing Spondylitis Disease Activity Index (BASDAI) scores compared to placebo at week 24. Improvements from baseline were maintained through week 56 in both studies.

Atopic dermatitis

The efficacy and safety of upadacitinib 15 mg and 30 mg once daily were assessed in three Phase 3 randomized, double-blind, multicenter studies (MEASURE UP 1, MEASURE UP 2 and AD UP) in a total of 2782 patients 12 years of age and older (Table 11). Upadacitinib was evaluated in 542 adolescent (344 in the primary analysis) and 2240 adult patients with moderate to severe atopic dermatitis (AD) not adequately controlled by topical medication(s). At baseline, patients had to have all the following: an Investigator's Global Assessment (vIGA-AD) score ≥3 in the overall assessment of AD (erythema, induration/papulation, and oozing/crusting) on an increasing severity scale of 0 to 4, an Eczema Area and Severity Index (EASI) score ≥16 (composite score assessing extent and severity of erythema, oedema/papulation, scratches and lichenification across 4 different body sites), a minimum body surface area (BSA) involvement of ≥10%, and weekly average Worst Pruritus Numerical Rating Scale (NRS) ≥4.

In all three studies, patients received upadacitinib once daily doses of 15 mg, 30 mg or matching placebo for 16 weeks. In the AD UP study, patients also received concomitant topical corticosteroids (TCS). Following completion of the double-blinded period, patients originally randomized to upadacitinib were to continue receiving the same dose until week 260. Patients in the placebo group were re-randomized in a 1:1 ratio to receive upadacitinib 15 mg or 30 mg until week 260.

Table 11. Clinical Trial Summary

Study Name	**Treatment Arms**	**Key Outcome Measures**
MEASURE UP 1 and MEASURE UP 2	• Upadacitinib 15 mg • Upadacitinib 30 mg • Placebo	Co-Primary Endpoints at Week 16: • EASI 75 • vIGA-AD 0/1 Key Secondary Endpoints (at Week 16 except where noted) • EASI 90/100 • EASI 75 at Week 2 • % change in EASI • % change in SCORAD • Worst Pruritus NRS improvement ≥ 4 at Week 1 and 16 • Worst Pruritus NRS improvement ≥ 4 at Day 2 (30 mg), Day 3 (15 mg) • % change in Worst Pruritus NRS • EASI increase ≥ 6.6 points (flare) during double-blind period • ADerm-SS TSS-7 improvement ≥ 28 • ADerm-SS Skin Pain improvement ≥ 4 • ADerm-IS Sleep improvement ≥ 12 • ADerm-IS Emotional State improvement ≥ 11 • ADerm-IS Daily Activities improvement ≥ 14 • POEM improvement ≥ 4 • HADS-A < 8 and HADS-D < 8 • DLQI 0/1 • DLQI improvement ≥ 4
AD UP	• Upadacitinib 15 mg + TCS • Upadacitinib 30 mg + TCS • Placebo + TCS	Co-Primary Endpoints at Week 16: • EASI 75 • vIGA-AD 0/1 Key Secondary Endpoints (at Week 16 except where noted) • EASI 75 at Week 2 and 4 • EASI 90 at Week 4 and 16 • EASI 100 (30 mg) • % change in EASI • Worst Pruritus NRS improvement ≥ 4 at Week 1, 4 and 16 • % change in Worst Pruritus NRS
Abbreviations: SCORAD = SCORing Atopic Dermatitis, POEM: Patient Oriented Eczema Measure, DLQI: Dermatology Life Quality Index, HADS: Hospital Anxiety and Depression Scale, ADerm-SS = Atopic Dermatitis Symptom Scale, ADerm-IS: Atopic Dermatitis Impact Scale		

Clinical Response

Monotherapy Studies (MEASURE UP 1 AND MEASURE UP 2)

In the MEASURE UP studies, a significantly greater proportion of patients treated with upadacitinib 15 mg or 30 mg achieved vIGA-AD 0 or 1 response and achieved EASI 75 compared to placebo at week 16 (Table 12). A rapid improvement in skin clearance (defined as EASI 75 by week 2) was achieved for both doses compared to placebo ($p < 0.001$).

A significantly greater proportion of patients treated with upadacitinib 15 mg or 30 mg achieved clinically meaningful improvement in itch (defined as a ≥ 4-point reduction in the Worst Pruritus NRS) compared to placebo at week 16. Rapid improvement in itch (defined as a ≥ 4-point reduction in Worst Pruritus NRS by week 1) was achieved for both doses compared to placebo ($p < 0.001$), with differences observed as early as 1 day after initiating upadacitinib 30 mg (Day 2, $p < 0.001$) and 2 days after initiating upadacitinib 15 mg (Day 3, $p < 0.001$).

A significantly smaller proportion of patients treated with upadacitinib 15 mg or 30 mg had a disease flare, defined as a clinically meaningful worsening of disease (increase in EASI by ≥ 6.6), during the initial 16 weeks of treatment compared to placebo ($p < 0.001$).

Figure 1 and Figure 2 show proportion of patients achieving an EASI 75 response and the proportion of patients with ≥4-point improvement in the Worst Pruritus NRS, respectively up to week 16.

Table 12: Efficacy results of upadacitinib monotherapy studies at week 16

Study	**MEASURE UP 1**			**MEASURE UP 2**		
Treatment Group	PBO	UPA 15 mg	UPA 30 mg	PBO	UPA 15 mg	UPA 30 mg
Number of subjects randomized	281	281	285	278	276	282
% responders						
vIGA-AD 0/1[a,b]	8.4	48.1[f]	62.0[f]	4.7	38.8[f]	52.0[f]
EASI 75[a]	16.3	69.6[f]	79.7[f]	13.3	60.1[f]	72.9[f]
EASI 90[a]	8.1	53.1[f]	65.8[f]	5.4	42.4[f]	58.5[f]
EASI 100[a]	1.8	16.7[f]	27.0[f]	0.7	14.1[f]	18.8[f]
Worst Pruritus NRS[c] *(≥ 4-point improvement)*	11.8 N=272	52.2[f] N=274	60.0[f] N=280	9.1 N=274	41.9[f] N=270	59.6[f] N=280
Worst Pruritus NRS 0 or 1[d]	5.5 N=275	36.6[g] N=279	47.5[g] N=282	4.3 N=277	26.9[g] N=275	44.1[g] N=281
Mean percent change (SE)[e]						
EASI	-40.7 (2.28)	-80.2[f] (1.91)	-87.7[f] (1.87)	-34.5 (2.59)	-74.1[f] (2.20)	-84.7[f] (2.18)
SCORAD	-32.7 (2.33)	-65.7[f] (1.78)	-73.1[f] (1.73)	-28.4 (2.50)	-57.9[f] (2.01)	-68.4[f] (2.04)
Worst Pruritus NRS	-26.1 (5.41)	-62.8[f] (4.49)	-72.0[f] (4.41)	-17.0 (2.73)	-51.2[f] (2.34)	-66.5[f] (2.31)

Abbreviations: UPA= upadacitinib (RINVOQ®); PBO = placebo
[a] Based on number of subjects randomized
[b] Responder was defined as a patient with vIGA-AD 0 or 1 (“clear” or “almost clear”) with a reduction of ≥ 2 points on a 0-4 ordinal scale
[c] N = number of patients whose baseline Worst Pruritus NRS is ≥ 4
[d] N = number of patients whose baseline Worst Pruritus NRS is > 1
[e] % change = least squares mean percent change relative to baseline
[f] multiplicity-controlled $p < 0.001$ upadacitinib vs placebo comparison
[g] nominal $p<0.001$ upadacitinib vs placebo comparison

Figure 1: Proportion of patients achieving an EASI 75 response in monotherapy studies

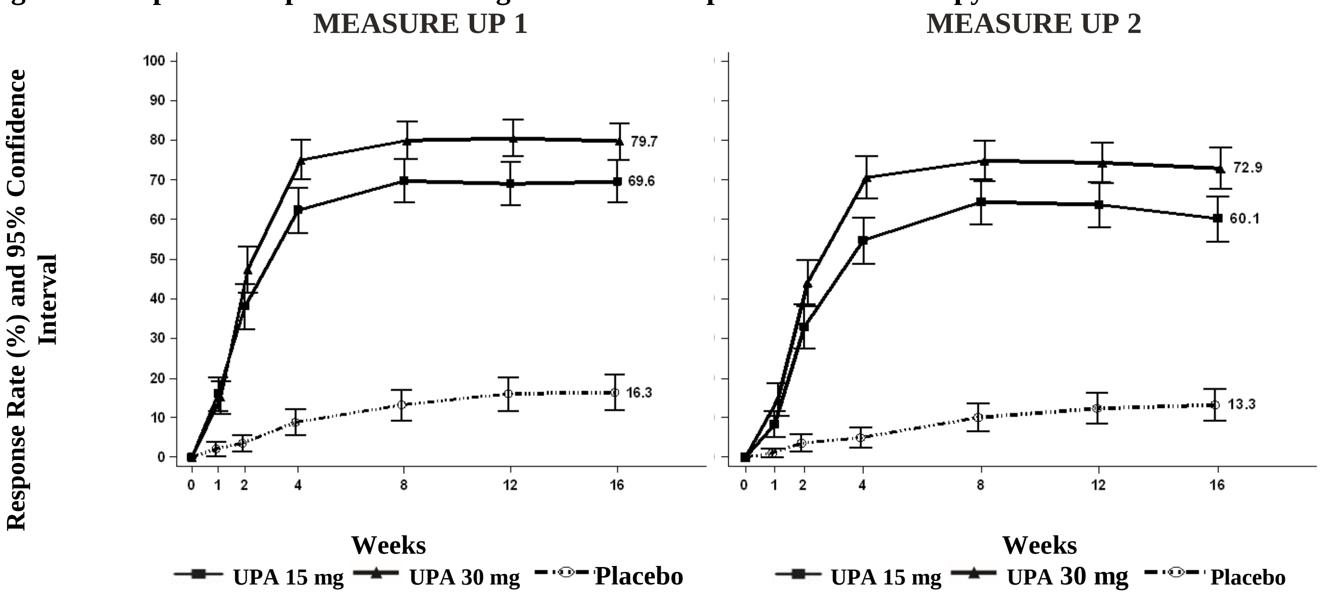
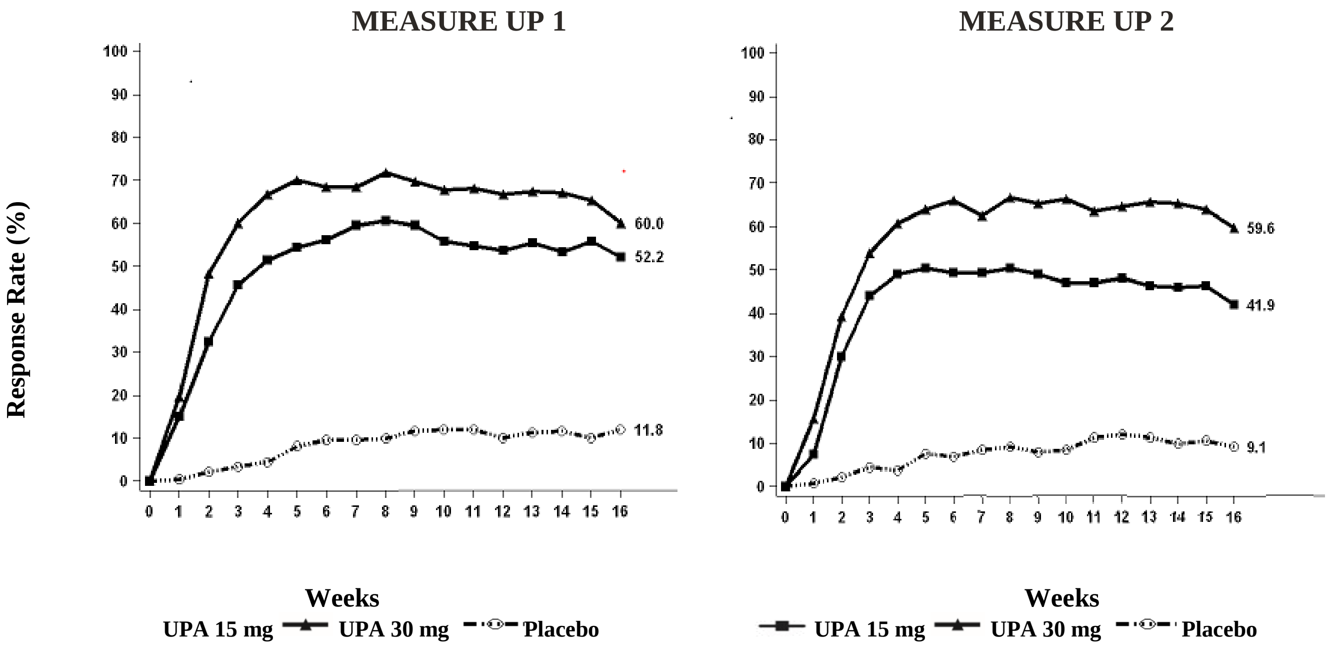


Figure 2: Proportion of patients with ≥4-point improvement in the Worst Pruritus NRS in monotherapy studies



In both studies, results at week 16 continued to be observed through week 52 in patients treated with upadacitinib 15 mg or 30 mg.

Treatment effects in subgroups (weight, age, gender, race, and prior systemic treatment with immunosuppressants) in both studies were consistent with the results in the overall study population.

Concomitant TCS Study (AD UP)

In AD UP, a significantly greater proportion of patients treated with upadacitinib 15 mg + TCS or 30 mg + TCS achieved vIGA-AD 0 or 1 response and achieved EASI 75 compared to placebo + TCS at week 16 (Table 12). A rapid improvement in skin clearance (defined as EASI 75 by week 2) was achieved for both doses compared to placebo + TCS ($p < 0.001$). In addition, a higher EASI 90 response rate was achieved at week 4 for both doses compared to placebo + TCS ($p < 0.001$).

A significantly greater proportion of patients treated with upadacitinib 15 mg + TCS or 30 mg + TCS achieved a clinically meaningful improvement in itch (defined as a ≥ 4-point reduction in the Worst Pruritus NRS) compared to placebo + TCS at week 16. A rapid improvement in itch (defined as a ≥ 4-point reduction in Worst Pruritus NRS by week 1) was achieved for both doses compared to placebo + TCS ($p < 0.001$).

Figure 3 and Figure 4 show proportion of patients achieving an EASI 75 response and the proportion of patients with ≥4-point improvement in the Worst Pruritus NRS, respectively up to week 16.

Table 13: Efficacy results of upadacitinib + concomitant TCS at week 16

Treatment Group	**Placebo + TCS**	**UPA 15 mg + TCS**	**UPA 30 mg + TCS**
Number of subjects randomized	304	300	297
% responders			
vIGA-AD 0/1[a,b]	10.9	39.6[f]	58.6[f]
EASI 75[a]	26.4	64.6[f]	77.1[f]
EASI 90[a]	13.2	42.8[f]	63.1[f]
EASI 100[a]	1.3	12.0[g]	22.6[f]
Worst Pruritus NRS[c] *(≥ 4-point improvement)*	15.0 N=294	51.7[f] N=288	63.9[f] N=291
Worst Pruritus NRS 0 or 1[d]	7.3 N=300	33.1[g] N=296	43.0[g] N=293
Mean percent change (SE)[e]			
EASI	-45.9 (2.16)	-78.0[f] (1.98)	-87.3[f] (1.98)
SCORAD	-33.6 (1.90)	-61.2[g] (1.70)	-71.0[g] (1.71)
Worst Pruritus NRS	-25.1 (3.35)	-58.1[f] (3.11)	-66.9[f] (3.12)

[a] Based on number of subjects randomized
[b] Responder was defined as a patient with vIGA-AD 0 or 1 (“clear” or “almost clear”) with a reduction of ≥ 2 points on a 0-4 ordinal scale
[c] N = number of patients whose baseline Worst Pruritus NRS is ≥ 4
[d] N = number of patients whose baseline Worst Pruritus NRS is > 1
[e] % change = least squares mean percent change relative to baseline
[f] multiplicity-controlled $p < 0.001$ upadacitinib + TCS vs placebo + TCS comparison
[g] nominal $p <0.001$upadacitinib + TCS vs placebo + TCS comparison

Figure 3: Proportion of patients achieving an EASI 75 response AD UP Study

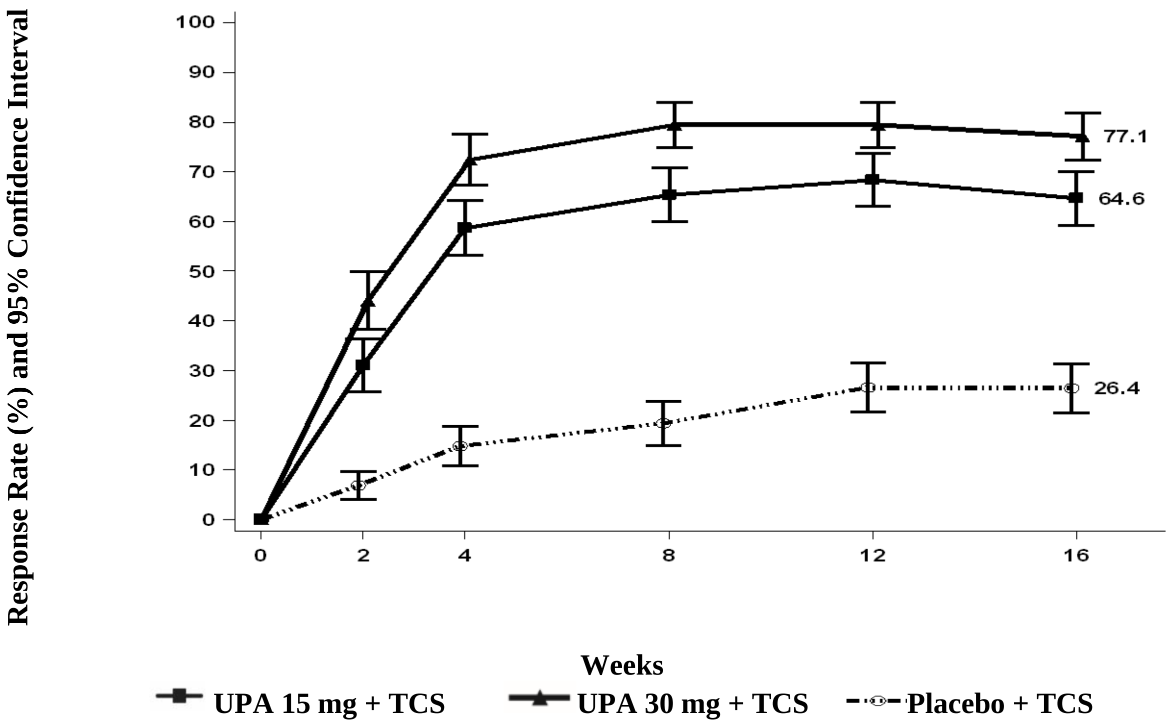
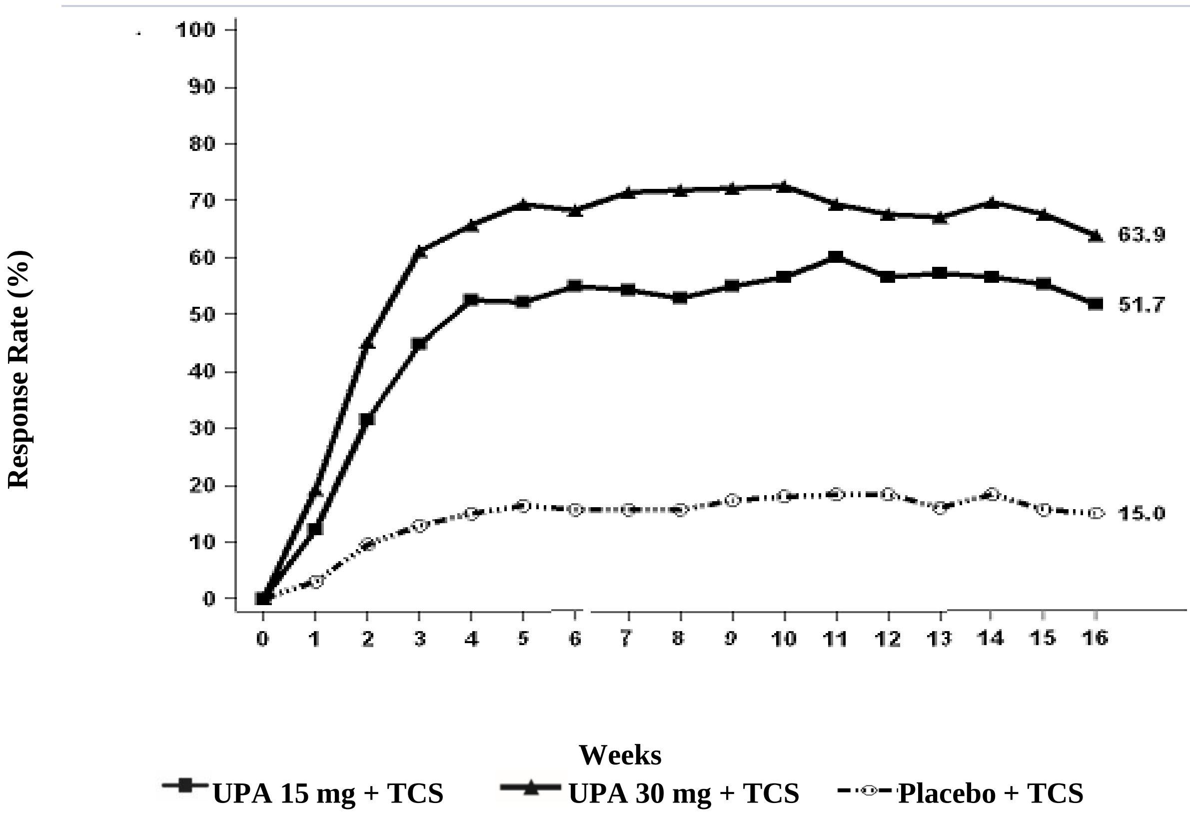


Figure 4: Proportion of patients with ≥4-point improvement in the Worst Pruritus NRS in AD UP Study



Treatment effects in subgroups (weight, age, gender, race, and prior systemic treatment with immunosuppressants) in AD UP were consistent with the results in the overall study population.

Subjects treated with either upadacitinib 15 mg or 30 mg had significantly more days free of TCS use with a concurrent EASI 75 response (mean: 33.5 and 47.5 days, respectively) over the 16-week period, compared to placebo group (mean: 7.9 days).

Results at week 16 continued to be observed through week 52 in patients treated with upadacitinib 15 mg or 30 mg.

Quality of Life/Patient reported outcomes

In the MEASURE UP studies, a significantly greater proportion of patients treated with upadacitinib 15 mg or 30 mg reported clinically meaningful reductions in the symptoms of AD and the impact of AD on health-related quality of life compared to placebo at week 16 (Table 14). A significantly greater proportion of patients treated with upadacitinib achieved clinically meaningful reductions in AD symptom severity as measured by ADerm-SS TSS-7 and ADerm-SS Skin Pain compared to placebo at week 16. A greater proportion of patients treated with upadacitinib achieved clinically meaningful reductions in the patient-reported effects of AD on sleep, daily activities and emotional state as measured by the ADerm-IS domain scores compared to placebo at week 16. Similarly, compared to placebo at week 16, a greater proportion of patients treated with upadacitinib achieved clinically meaningful improvements in AD symptom frequency and health-related quality of life as measured by the POEM and DLQI.

Anxiety and depression symptoms as measured by the HADS score were significantly reduced; in patients with baseline HADS-anxiety or HADS-depression subscale scores ≥ 8 (the cut-off value for anxiety or depression), a greater proportion of patients in the upadacitinib 15 mg or 30 mg groups achieved HADS-anxiety and HADS-depression scores < 8 at week 16 compared to placebo (Table 14).

Table 14: Patient-reported outcomes results of upadacitinib monotherapy studies at week 16

Study	**MEASURE UP 1**			**MEASURE UP 2**		
Treatment group	PBO	UPA 15 mg	UPA 30 mg	PBO	UPA 15 mg	UPA 30 mg
Number of subjects randomized	281	281	285	278	276	282
% responders						
ADerm-SS TSS-7 *(≥ 28-point improvement)* [a,b]	15.0 N=226	53.6[h] N=233	67.9[h] N=246	12.7 N=244	53.0[h] N=230	66.2[h] N=234
ADerm-SS Skin Pain *(≥ 4-point improvement)*[a]	15.0 N=233	53.6[h] N=237	63.5[h] N=249	13.4 N=247	49.4[h] N=237	65.1[h] N=238
ADerm-IS Sleep *(≥ 12-point improvement)* [a,c]	13.2 N=220	55.0[h] N=218	66.1[h] N=218	12.4 N=233	50.2[h] N=219	62.3[h] N=228
ADerm-IS Daily Activities *(≥ 14-point improvement)*[a,d]	20.3 N=197	65.0[h] N=203	73.2[h] N=205	18.9 N=227	57.0[h] N=207	69.5[h] N=223
ADerm-IS Emotional State	19.8 N=212	62.6[h] N=227	72.6[h] N=226	16.7 N=234	57.0[h] N=228	71.5[h] N=228

(≥ 11-point improvement) [a,e]						
DLQI *(DLQI 0/1)*[f]	4.4 N=252	30.3[h] N=258	41.5[h] N=261	4.7 N=257	23.8[h] N=252	37.9[h] N=256
DLQI *(≥ 4-point improvement)* [a]	29.0 N=250	75.4[h] N=254	82.0[h] N=256	28.4 N=250	71.7[h] N=251	77.6[h] N=251
POEM *(≥ 4-point improvement)* [a]	22.8 N=276	75.0[h] N=278	81.4[h] N=280	28.7 N=268	70.9[h] N=268	83.5[h] N=269
HADS *(HADS-A < 8 and HADS-D < 8)* [g]	14.3 N=126	45.5[h] N=145	49.2[h] N=144	11.4 N=140	46.0[h] N=137	56.1[h] N=146

The threshold values specified correspond to the minimal clinically important difference (MCID) and was used to determine response.
[a] N = number of patients whose baseline score is greater than or equal to the MCID.
[b] ADerm-SS TSS-7 assesses itch while asleep, itch while awake, skin pain, skin cracking, pain caused by skin cracking, dry skin, and flaking due to AD.
[c] ADerm-IS Sleep assesses difficulty falling asleep, sleep impact, and waking up at night due to AD.
[d] ADerm-IS Daily Activities assesses AD's effect on household activities, physical activities, social activities, and concentration.
[e] ADerm-IS Emotional State assesses self-consciousness, embarrassment, and sadness due to AD.
[f] N = number of patients whose baseline DLQI score is > 1.
[g] N = number of patients whose baseline HADS-A or HADS-D is ≥ 8.
[h] multiplicity-controlled $p < 0.001$ upadacitinib vs placebo comparison.

Adolescent population

A total of 542 adolescents aged 12 to 17 years with moderate to severe atopic dermatitis were randomized across the three Phase 3 studies, of which 344 were evaluated for the primary analysis. Adolescents in the primary analysis were randomized to either 15 mg (N=114) or 30 mg (N=114) upadacitinib or matching placebo (N=116), in monotherapy or combination with topical corticosteroids. Efficacy was consistent between the adolescents and adults (Table 15). The adverse event profile in adolescents was generally similar to that in adults. Safety and efficacy of upadacitinib in adolescents weighing less than 40 kg and in patients less than 12 years of age with atopic dermatitis have not been established.

Table 15: Efficacy results of upadacitinib for adolescents at week 16

Study	**MEASURE UP 1**			**MEASURE UP 2**			**AD UP**		
Treatment Group	PBO	UPA 15 mg	UPA 30 mg	PBO	UPA 15 mg	UPA 30 mg	PBO + TCS	UPA 15 mg + TCS	UPA 30 mg + TCS
Number of adolescent subjects randomized	40	42	42	36	33	35	40	39	37
% responders									
vIGA-AD 0/1[a,b]	7.5%	38.1%	69.0%	2.8%	42.4%	62.5%	7.5%	30.8%	64.9%
EASI 75[a]	8.3%	71.4%	83.3%	13.9%	66.7%	74.5%	30.0%	56.4%	75.7%

Worst Pruritus NRS[c] *(≥ 4-point improvement)*	15.4% N=39	45.0% N=40	54.8% N=42	2.8% N=36	33.3% N=30	50.0% N=34	13.2% N=38	41.7% N=36	54.5% N=33

[a] Based on number of subjects randomized
[b] Responder was defined as a patient with vIGA-AD 0 or 1 ("clear" or "almost clear") with a reduction of ≥ 2 points on a 0-4 ordinal scale
[c] N = number of patients whose baseline Worst Pruritus NRS is ≥ 4

Results at Week 16 were maintained through Week 76 in adolescents treated with RINVOQ 15 mg or 30 mg.

The European Medicines Agency has deferred the obligation to submit the results of studies with RINVOQ® in one or more subsets of the paediatric population in chronic idiopathic arthritis (including rheumatoid arthritis, psoriatic arthritis, spondyloarthritis, juvenile idiopathic arthritis and atopic dermatitis) (see **Posology and method of administration** section for information on paediatric use).

Non-radiographic Axial Spondyloarthritis

The efficacy and safety of RINVOQ 15 mg once daily were assessed in a randomized, double-blind, multicenter, placebo-controlled study in patients 18 years of age or older with active non-radiographic axial spondyloarthritis based upon the Bath Ankylosing Spondylitis Disease Activity Index (BASDAI) ≥ 4, Patient's Assessment of Total Back Pain score ≥ 4, and objective signs of inflammation (Table 16). The study included a long-term extension for up to 2 years.

Table 16. Clinical Trial Summary

Study Name	**Population (n)[a]**	**Treatment Arms**	**Key Outcome Measures**
SELECT-AXIS 2	NSAID-IR[b,c] (314)	• Upadacitinib 15 mg • Placebo	Primary Endpoint: • ASAS40 at Week 14 Key Secondary Endpoints at Week 14: • ASDAS-CRP • SPARCC MRI score (SI joints) • BASDAI 50 • ASDAS Inactive Disease • Total Back Pain • Nocturnal Back Pain • ASDAS Low Disease Activity • ASAS Partial Remission • BASFI (function) • AS Quality of Life • ASAS Health Index • ASAS20 • BASMI (spinal mobility) • MASES (enthesitis)

Abbreviations: ASAS40 = Assessment of SpondyloArthritis international Society ≥40% improvement; ASDAS-CRP = Ankylosing Spondylitis Disease Activity Score C-Reactive Protein; BASDAI = Bath Ankylosing Spondylitis Disease Activity Index; BASFI = Bath Ankylosing Spondylitis Functional Index; NSAID = Nonsteroidal Anti-inflammatory Drug; SPARCC MRI = Spondyloarthritis Research Consortium of Canada Magnetic Resonance Imaging
[a] Patients must have had objective signs of inflammation indicated by elevated C-reactive protein (CRP) (defined as > upper limit of normal), and/or sacroiliitis on magnetic resonance imaging (MRI),

and no definitive radiographic evidence of structural damage on sacroiliac joints. [b] Patients who had an inadequate response to at least two NSAIDs or had intolerance to or contraindication for NSAIDs [c] At baseline, 29.1% of the patients were on a concomitant csDMARD and 32.9% of the patients had an inadequate response or intolerance to bDMARD therapy.

Clinical Response

In SELECT-AXIS 2, a significantly greater proportion of patients treated with RINVOQ 15 mg achieved an ASAS40 response compared to placebo at Week 14 (Table 17, Figure 5). Time to onset of efficacy was rapid across measures with greater responses seen as early as Week 2 for ASAS40.

Treatment with RINVOQ 15 mg resulted in greater improvement in individual ASAS components (patient global assessment of disease activity, total back pain assessment, inflammation, and function) and other measures of disease activity, including BASDAI compared to placebo at Week 14.

The efficacy of RINVOQ 15 mg was demonstrated across subgroups including gender, baseline BMI, symptom duration of non-radiographic axial spondyloarthritis, baseline hsCRP, MRI sacroiliitis, and prior use of bDMARDs.

Figure 5. Percent of Patients Achieving ASAS40

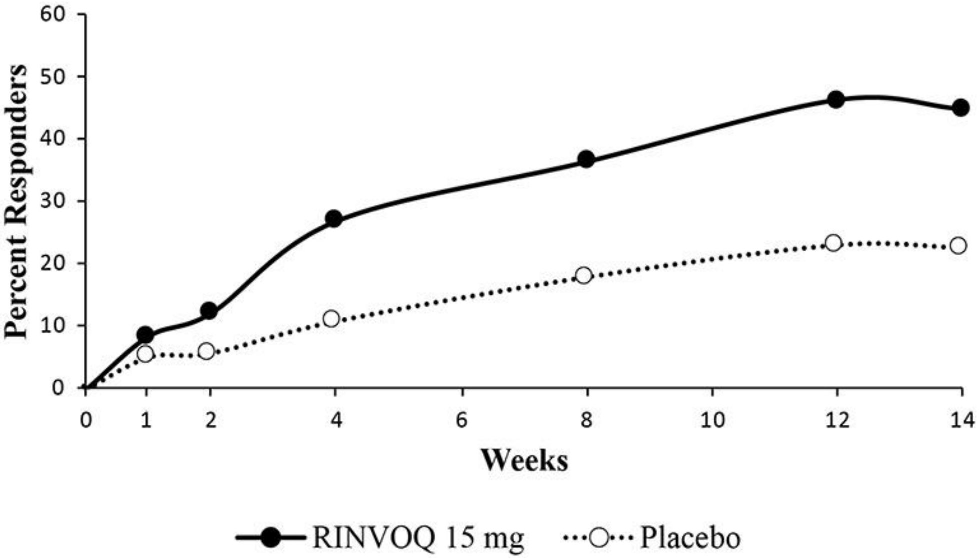


Table 17. Clinical Response

	PBO (N=157)	UPA 15 mg (N=156)
ASAS40 (%)		
Week 14	22.5	44.9[a]
Week 52	42.7	62.8[c]
ASAS20 (%)		
Week 14	43.8	66.7[a]
Week 52	52.2	68.6[d]
ASAS Partial Remission (%)		
Week 14	7.6	18.6[b]
Week 52	17.8	35.3[c]
BASDAI 50 (%)		
Week 14	22.1	42.3[a]
Week 52	40.1	55.8[d]
ASDAS-CRP (Change from Baseline)		

Week 14	-0.71	-1.36[a]
Week 52	-1.23	-1.80[c]
ASDAS Inactive Disease (%)		
Week 14	5.2	14.1[b]
Week 52	10.8	32.7[c]
ASDAS Low Disease Activity (%)		
Week 14	18.3	42.3[a]
Week 52	32.5	55.8[c]
ASDAS Major Improvement (%)		
Week 14	8.5	23.7[c]
Week 52	20.4	37.8[c]
hsCRP mg/L (Change from Baseline)		
Week 14	-1.45	-6.50[c]
Week 52	-2.62	-6.91[c]

[a] multiplicity-controlled p≤0.001 upadacitinib vs placebo comparison
[b] multiplicity-controlled p≤0.01 upadacitinib vs placebo comparison
[c] nominal p≤0.001 upadacitinib vs placebo comparison
[d] nominal p≤0.01 upadacitinib vs placebo comparison
For binary endpoints, results are based on non-responder imputation in conjunction with multiple imputation. For continuous endpoints, results are based on the least squares mean change from baseline using mixed models for repeated measures analysis.

Efficacy was maintained through 2 years as assessed by the endpoints presented in Table 17.

Table 18. Components of ASAS Response (mean change from baseline)

Treatment Group	**PBO (N=157)**	**UPA 15 mg (N=156)**
Patient Global Assessment of Disease Activity[a]		
Week 14	-1.87	-2.89[d]
Week 52	-3.30	-4.27[d]
Total Back Pain[a]		
Week 14	-2.00	-2.91[c]
Week 52	-3.46	-4.22[e]
BASFI[a]		
Week 14	-1.47	-2.61[c]
Week 52	-2.74	-3.71[d]
Inflammation[b]		
Week 14	-1.93	-3.05[d]
Week 52	-3.38	-4.03[e]

Results are based on the least squares mean change from baseline using mixed models for repeated measures analysis
[a] Numeric rating scale (NRS): 0 = best, 10 = worst
[b] mean of BASDAI questions 5 and 6 assessing morning stiffness severity and duration: 0 = best, 10 = worst
[c] multiplicity-controlled p≤0.001 upadacitinib vs placebo comparison
[d] nominal p≤0.001 upadacitinib vs placebo comparison
[e] nominal p≤0.05 upadacitinib vs placebo comparison

Physical Function Response and Health-Related Outcomes
Patients treated with RINVOQ 15 mg showed significant improvement in physical function from baseline compared to placebo as assessed by the BASFI at Week 14 (Table 18).

Patients treated with RINVOQ 15 mg showed significant improvements in total back pain and nocturnal back pain compared to placebo at Week 14. These improvements were observed as early as Week 2 for total back pain and Week 4 for nocturnal back pain.

Patients treated with RINVOQ 15 mg showed significant improvements in health-related quality of life and overall health as measured by Ankylosing Spondylitis Quality of Life (ASQoL) and ASAS Health Index, respectively, compared to placebo at Week 14.

Patients treated with RINVOQ 15 mg experienced greater improvement from baseline in fatigue as measured by FACIT-F score compared to placebo at Week 14.

Improvements in BASFI, total and nocturnal back pain, ASQoL, ASAS Health Index, and FACIT-F were maintained through 2 years.

Enthesitis

Patients with pre-existing enthesitis treated with RINVOQ 15 mg showed greater improvement in enthesitis compared to placebo as measured by change from baseline in Maastricht Ankylosing Spondylitis Enthesitis Score (MASES) at Week 14.

Improvement in enthesitis was maintained through 2 years.

Objective Measures of Inflammation

Signs of inflammation were assessed by MRI and expressed as change from baseline in the SPARCC score. Improvement of inflammatory signs in both sacroiliac joints and the spine was observed in patients treated with upadacitinib 15 mg. At Week 14, significant improvement of inflammatory signs in the sacroiliac joints was observed in patients treated with upadacitinib 15 mg compared to placebo.

Improvement in inflammation as assessed by MRI was maintained through 2 years.

Ankylosing Spondylitis (AS, radiographic axial spondyloarthritis)

The efficacy and safety of RINVOQ 15 mg once daily were assessed in two randomised, double-blind, multicentre, placebo-controlled studies in patients 18 years of age or older with active ankylosing spondylitis based upon the Bath Ankylosing Spondylitis Disease Activity Index (BASDAI) ≥4 and Patient’s Assessment of Total Back Pain score ≥4. Both studies included long-term extensions for up to 2 years.

SELECT-AXIS 1 was a 14-week placebo-controlled trial in 187 ankylosing spondylitis patients with an inadequate response to at least two NSAIDs or intolerance to or contraindication for NSAIDs and had no previous exposure to biologic DMARDs. At baseline, patients had symptoms of ankylosing spondylitis for an average of 14.4 years and approximately 16% of the patients were on a concomitant csDMARD. Patients received upadacitinib 15 mg once daily or placebo. At week 14, all patients randomised to placebo were switched to upadacitinib 15 mg once daily. The primary endpoint was the proportion of patients achieving an Assessment of SpondyloArthritis international Society 40 (ASAS40) response at week 14.

SELECT-AXIS 2 (AS) was a 14-week placebo-controlled trial in 420 ankylosing spondylitis patients with prior exposure to bDMARDs (77.4% had lack of efficacy to either a tumour necrosis factor (TNF) blocker or interleukin-17 inhibitor (IL-17i); 30.2% had intolerance; 12.9% had prior exposure but not lack of efficacy to two bDMARDs). At baseline, patients had symptoms of ankylosing spondylitis for an average of 12.8 years and approximately 31% of the patients were on a concomitant csDMARD. Patients received upadacitinib 15 mg once daily or placebo. At week 14, all patients randomised to placebo were switched to upadacitinib 15 mg once daily. The primary endpoint was the proportion of patients achieving an Assessment of SpondyloArthritis international Society 40 (ASAS40) response at week 14.

Clinical Response

In both studies, a significantly greater proportion of patients treated with RINVOQ 15 mg achieved an ASAS40 response compared to placebo at Week 14 (Table 19). A numerical difference between treatment groups was observed from week 2 in SELECT-AXIS 1 and week 4 in SELECT-AXIS 2 (AS) for ASAS40.

Treatment with RINVOQ 15 mg resulted in improvements in individual ASAS components, (patient global assessment of disease activity, total back pain assessment, inflammation, and function) and other measures of disease activity, including hsCRP, at week 14 compared to placebo.

The efficacy of upadacitinib 15 mg was demonstrated regardless of subgroups evaluated including gender, baseline BMI, symptom duration of AS, baseline hsCRP, and prior use of bDMARDs.

Table 19. Clinical Response

Study	**SELECT-AXIS 1 bDMARD-naïve**		**SELECT-AXIS 2 bDMARD-IR**	
Treatment Group	**PBO**	**UPA 15 mg**	**PBO**	**UPA 15 mg**
N	94	93	209	211
ASAS40 (% of patients)				
Week 14	25.5	51.6[a]	18.2	44.5[a]
Week 52		80.2		73.7
Week 104		85.9		81.5
ASAS20 (% of patients)				
Week 14	40.4	64.5[c]	38.3	65.4[a]
Week 52		87.7		88.7
Week 104		90.1		91.1
ASAS Partial Remission (% of patients)				
Week 14	1.1	19.4[a]	4.3	17.5[a]
Week 52		50.0		33.5
Week 104		51.4		42.9
BASDAI 50 (% of patients)				
Week 14	23.4	45.2[b]	16.7	43.1[a]
Week 52		77.8		64.9
Week 104		88.7		78.6
ASDAS-CRP (Change from Baseline)				
Week 14	-0.54	-1.45[a]	-0.49	-1.52[a]
Week 52		-2.05		-2.01
Week 104		-2.10		-2.23
ASDAS Inactive Disease (% of patients)				
Week 14	0	16.1[c]	1.9	12.8[a]
Week 52		46.2		30.4
Week 104		45.6		38.0
ASDAS Low Disease Activity (% of patients)[d]				
Week 14	10.6	49.5[c]	10.1	44.1[a]
Week 52		85.9		65.8
Week 104		86.8		71.1

ASDAS Major Improvement (% of patients)				
Week 14	5.3	32.3[c]	4.8	30.3[c]
Week 52		55.8		52.7
Week 104		55.2		62.0
hsCRP mg/L (Change from Baseline)				
Week 14	0.18	-8.20[c]	0.43	-10.90[c]
Week 52		-7.29		-9.45
Week 104		-8.03		-10.92

[a] multiplicity-controlled $p \leq 0.001$ upadacitinib vs placebo comparison
[b] multiplicity-controlled $p \leq 0.01$ upadacitinib vs placebo comparison
[c] nominal $p \leq 0.001$ upadacitinib vs placebo comparison
[d] post-hoc analysis for SELECT-AXIS 1; multiplicity-controlled endpoint in SELECT-AXIS 2.
For binary endpoints, Week 14 results are based on non-responder imputation (SELECT-AXIS 1) and on non-responder imputation in conjunction with multiple imputation (SELECT-AXIS 2). For continuous endpoints, Week 14 results are based on the least squares mean change from baseline using mixed models for repeated measures analysis. For binary and continuous endpoints, Week 52 and Week 104 results are based on as-observed data.

In both studies, efficacy was maintained through 2 years as assessed by the endpoints presented in Table 19.

Table 20. Components of ASAS Response (mean change from baseline)

Study	**SELECT-AXIS 1 bDMARD-naïve**		**SELECT-AXIS 2 bDMARD-IR**	
Treatment Group	**PBO**	**UPA 15 mg**	**PBO**	**UPA 15 mg**
N	94	93	209	211
Patient Global Assessment of Disease Activity[a]				
Week 14	-1.31	-2.96[d]	-1.38	-2.97[d]
Week 52		-4.54		-4.62
Week 104		-4.68		-5.14
Total Back Pain[a]				
Week 14	-1.68	-3.21[d]	-1.47	-3.00[c]
Week 52		-4.75		-4.60
Week 104		-4.79		-5.08
BASFI[a]				
Week 14	-1.30	-2.29[c]	-1.09	-2.26[c]
Week 52		-3.71		-3.68
Week 104		-3.76		-4.02
Inflammation[b]				
Week 14	-1.90	-3.15[d]	-1.59	-2.94[d]
Week 52		-4.80		-4.30
Week 104		-4.89		-4.72

Week 14 results are based on the least squares mean change from baseline using mixed models for repeated measures analysis; Week 52 and Week 104 results are based on as-observed data.
[a] Numeric rating scale (NRS): 0 = best, 10 = worst
[b] mean of BASDAI questions 5 and 6 assessing morning stiffness severity and duration: 0 = best, 10 = worst
[c] multiplicity-controlled $p \leq 0.001$ upadacitinib vs placebo comparison
[d] nominal $p \leq 0.001$ upadacitinib vs placebo comparison

Physical Function Response and Health-Related Outcomes
In both studies, patients treated with RINVOQ 15 mg showed significant improvement in physical function from baseline compared to placebo as assessed by the BASFI at Week 14 (Table 20).

In SELECT-AXIS 1, patients treated with RINVOQ 15 mg showed greater improvement in back pain as assessed by the Total Back Pain component of ASAS response and nocturnal back pain compared to placebo at Week 14. Improvements in total and nocturnal back pain were observed as early as Week 2.

In SELECT-AXIS 2, patients treated with RINVOQ 15 mg showed significant improvements in total back pain and nocturnal back pain compared to placebo at Week 14. These improvements were observed as early as Week 1 for total back pain and Week 2 for nocturnal back pain.

In both studies, improvement in the overall level of neck, back, or hip pain was demonstrated using BASDAI Question 2 and for peripheral pain and swelling (assessed by BASDAI question 3 on overall pain in joints other than in the neck, back, or hips).

In both studies, improvements in BASFI and pain were maintained through 2 years for patients receiving RINVOQ 15 mg.

In SELECT-AXIS 2, patients treated with RINVOQ 15 mg showed significant improvements in health-related quality of life and overall health as measured by ASQoL and ASAS Health Index, respectively, compared to placebo at Week 14. Improvements in ASQoL and ASAS Health Index were also observed in SELECT-AXIS 1 compared to placebo at Week 14. Improvements in ASQoL and ASAS Health Index were maintained through 2 years.

In SELECT-AXIS 2, patients treated with RINVOQ 15 mg experienced greater improvement from baseline in fatigue as measured by FACIT-F score compared to placebo at Week 14. Improvement in FACIT-F was maintained through 2 years.

Enthesitis
In SELECT-AXIS 2, patients with pre-existing enthesitis (n=310) treated with RINVOQ 15 mg showed significant improvement in enthesitis compared to placebo as measured by change from baseline in Maastricht Ankylosing Spondylitis Enthesitis Score (MASES) at week 14. Improvements in MASES were also observed in SELECT-AXIS 1 compared to placebo at Week 14. Improvement in enthesitis was maintained through 2 years.

Spinal mobility
In SELECT-AXIS 2 (AS), patients treated with RINVOQ 15 mg showed significant improvement in spinal mobility compared to placebo as measured by change from baseline in Bath Ankylosing Spondylitis Metrology Index (BASMI) at Week 14. Improvements in BASMI were also observed in SELECT-AXIS 1 compared to placebo at Week 14. Improvement in BASMI was maintained through 2 years.

Objective Measures of Inflammation
Signs of inflammation were assessed by MRI and expressed as change from baseline in the SPARCC score for spine and sacroiliac joints. In both studies, at Week 14, significant improvement of inflammatory signs in the spine was observed in patients treated with RINVOQ 15 mg compared to placebo. Additionally, patients treated with RINVOQ 15 mg demonstrated greater improvement of inflammatory signs in sacroiliac joints compared to placebo. Improvement in inflammation as assessed by MRI was maintained through 2 years.

Giant cell arteritis

The efficacy and safety of upadacitinib 15 mg once daily were assessed in SELECT-GCA, a Phase 3 randomised, double-blind, multicentre, placebo-controlled study in patients 50 years of age and older with new onset or relapsing giant cell arteritis. SELECT-GCA was a 52-week study in which 428 patients were randomised in a 2:1:1 ratio and dosed once daily with upadacitinib 15 mg, upadacitinib 7.5 mg, or placebo. All patients received background corticosteroid (prednisone or prednisolone) therapy. The upadacitinib-treated groups followed a pre-specified corticosteroid taper regimen with the aim to reach 0 mg by 26 weeks; the placebo-treated group followed a pre-specified corticosteroid taper regimen with the aim to reach 0 mg by 52 weeks. The primary endpoint was the proportion of patients achieving sustained remission at week 52 as defined by the absence of giant cell arteritis signs and symptoms from week 12 through week 52 and adherence to the protocol-defined corticosteroid taper regimen. Patients who prematurely discontinued study treatment (upadacitinib or placebo) or had a missing assessment were classified as non-responders. The study included a 52-week extension for a total study duration of up to 2 years.

Clinical response

Upadacitinib 15 mg and a 26-week corticosteroid taper showed superiority in achieving corticosteroid-free sustained remission at week 52 compared to placebo and a 52-week corticosteroid taper (Table 21). Results for each component of sustained remission and sustained complete remission at week 52 were consistent with those of the composite endpoints. For sustained remission at week 52 (the primary endpoint), a similar percentage of patients in each arm were classified as non-responders due to premature discontinuation of study treatment (placebo: 19.6%; upadacitinib 15 mg: 20.1%) or due to a missing assessment (placebo: 0.9%; upadacitinib 15 mg: 0.5%).

Treatment effects in subgroups (gender, age, race, prior use of interleukin-6 inhibitor, new onset or relapsing giant cell arteritis, baseline corticosteroid dose, and giant cell arteritis with or without polymyalgia rheumatica) were consistent with the results in the overall study population.

A significantly lower proportion of patients treated with upadacitinib 15 mg and a 26-week corticosteroid taper experienced at least one giant cell arteritis flare compared to those treated with placebo and a 52-week corticosteroid taper through week 52. In addition, the risk of flare in the upadacitinib arm was significantly lower compared to the placebo arm as measured by time to first flare through week 52 (Table 21).

Table 21. Clinical response in SELECT-GCA

Treatment Group	**PBO + 52-week corticosteroid taper N=112**	**UPA 15 mg + 26-week corticosteroid taper N=209**	**Treatment Difference (95% CI)**
Sustained remission at Week 52[a]	29.0%	46.4%	17.1%[e] (6.3, 27.8)
Sustained complete remission at Week 52[b]	16.1%	37.1%	20.7%[f] (11.3, 30.2)
Complete remission at Week 52[c]	19.6%	50.2%	30.3%[f] (20.4, 40.2)
Complete remission at Week 24[c]	36.1%	57.2%	20.8%[f] (9.7, 31.9)
Time to first GCA flare through Week 52[d]			0.57[e,g] (0.399, 0.826)

Patients with one or more GCA flares through Week 52[d]	55.6%	34.3%	0.47[e,h] (0.29, 0.74)
Abbreviations: ESR = erythrocyte sedimentation rate; GCA = giant cell arteritis; hsCRP = high sensitivity C-reactive protein; PBO = placebo; UPA = upadacitinib [a] Sustained remission is defined as having achieved both the absence of GCA signs and symptoms from Week 12 through Week 52 and adherence to the protocol-defined corticosteroid taper regimen [b] Sustained complete remission is defined as having achieved absence of GCA signs and symptoms from Week 12 through Week 52, normalization of ESR (to ≤ 30 mm/hr; if ESR > 30 mm/hr and elevation is not attributable to GCA, this criterion can still be met) from Week 12 through Week 52, normalization of hsCRP to < 1 mg/dL without elevation to ≥ 1 mg/dL (on 2 consecutive visits) from Week 12 through Week 52, and adherence to the protocol-defined corticosteroid taper regimen [c] Complete remission is defined as having achieved absence of GCA signs and symptoms, normalization of ESR (to ≤ 30 mm/hr; if ESR > 30 mm/hr and elevation is not attributable to GCA, this criterion can still be met), normalization of hsCRP to < 1 mg/dL, and adherence to the protocol-defined corticosteroid taper regimen [d] GCA flare is defined as an event representing recurrence of GCA signs or symptoms or an ESR measurement > 30 mm/hr (attributable to GCA) and requiring an increase in corticosteroid dose, and is only considered after all of the 3 following criteria are met: absence of recurrence of GCA signs and symptoms, normalization of ESR, and no corticosteroid dose increase. Subjects who do not have an assessment that meets all 3 criteria are considered as having a GCA flare at baseline. Time to first GCA flare is calculated from the time when all three criteria above are met. Subjects who meet all 3 criteria above but never experience GCA flare are censored at the last assessment [e] p≤0.01 [f] p≤0.001 [g] Hazard ratio [h] Odds ratio			

Cumulative corticosteroid dose

Among patients who completed 52 weeks of follow-up, the cumulative corticosteroid exposure at week 52 was significantly lower in patients treated with upadacitinib 15 mg and a 26-week corticosteroid taper compared to placebo and a 52-week corticosteroid taper (median 1615 mg vs 2882 mg, respectively). The comparison of cumulative corticosteroid dose between the upadacitinib arm and the placebo arm is affected by the different pre-specified regimens for steroid tapering in the upadacitinib arm versus the placebo arm.

Health-related outcomes

Fatigue was assessed using FACIT-Fatigue score. Patients treated with upadacitinib 15 mg and a 26-week corticosteroid taper experienced significantly greater improvement from baseline compared to placebo and a 52-week corticosteroid taper in FACIT-Fatigue score at week 52 (4.0, 95% CI: 1.33, 6.76).

Health-related quality of life was assessed using SF-36. Patients receiving upadacitinib 15 mg and a 26-week corticosteroid taper experienced significantly greater improvement from baseline compared to placebo and a 52-week corticosteroid taper in the Physical Component Summary score of SF-36 at week 52 (3.75, 95% CI: 1.39, 6.11).

Ulcerative Colitis

Ulcerative colitis

The efficacy and safety of upadacitinib was evaluated in three multicentre, double-blind, placebo-controlled Phase 3 clinical studies: two replicate induction studies, UC-1 (U-ACHIEVE Induction) and UC-2 (U-ACCOMPLISH), and a maintenance study UC-3 (U-ACHIEVE

Maintenance). In addition, efficacy and safety of RINVOQ were assessed in a long-term extension study, UC-4.

Disease activity was based on the adapted Mayo score (aMS, Mayo scoring system excluding Physician's Global Assessment), which ranged from 0 to 9 and has three subscores that were each scored 0 (normal) to 3 (most severe): stool frequency subscore (SFS), rectal bleeding subscore (RBS) and a centrally-reviewed endoscopy subscore (ES).

Induction studies (UC-1 and UC-2)
In UC-1 and UC-2, 988 patients (473 and 515 patients, respectively) were randomised to upadacitinib 45 mg once daily or placebo for 8 weeks with a 2:1 treatment allocation ratio and included in the efficacy analysis. All enrolled patients had moderately to severely active ulcerative colitis defined as an aMS of 5 to 9 with an ES of 2 or 3 and demonstrated prior treatment failure including inadequate response, loss of response, or intolerance to prior conventional and/or biologic treatment. Prior treatment failure to at least 1 biologic therapy (prior biologic failure) was seen in 52% (246/473) and 51% (262/515) of patients, respectively. Previous treatment failure to conventional therapy but not biologics (without prior biologic failure) was seen in 48% (227/473) and 49% (253/515) of patients, respectively.

At baseline in UC-1 and UC-2, 39% and 37% of patients received corticosteroids, 1.1% and 0.6% of patients received methotrexate and 68% and 69% of patients received aminosalicylates. Concomitant use of thiopurine was not allowed during the studies. Patient disease activity was moderate (aMS ≥5, ≤7) in 61% and 60% of patients and severe (aMS >7) in 39% and 40% of patients.

The primary endpoint was clinical remission per aMS at week 8. Table 22 shows the primary and key secondary endpoints including clinical response, mucosal healing, histologic-endoscopic mucosal healing and deep mucosal healing.

Table 22. Proportion of patients meeting primary and key secondary efficacy endpoints at week 8 in the induction studies UC-1 and UC-2

	UC-1 (U-ACHIEVE)			**UC-2 (U-ACCOMPLISH)**		
Endpoint	**PBO N=154**	**UPA 45 mg N=319**	**Treatment Difference (95% CI)**	**PBO N=174**	**UPA 45 mg N=341**	**Treatment Difference (95% CI)**
Clinical remission[a]	4.8%	26.1%	21.6%* (15.8, 27.4)	4.1%	33.5%	29.0%* (23.2, 34.7)
Prior biologic failure[+]	0.4%	17.9%	17.5%	2.4%	29.6%	27.1%
Without prior biologic failure[+]	9.2%	35.2%	26.0%	5.9%	37.5%	31.6%
Clinical response[b]	27.3%	72.6%	46.3%* (38.4, 54.2)	25.4%	74.5%	49.4%* (41.7, 57.1)
Prior biologic failure[+]	12.8%	64.4%	51.6%	19.3%	69.4%	50.1%
Without prior biologic failure[+]	42.1%	81.8%	39.7%	31.8%	79.8%	48.0%
Mucosal healing[c]	7.4%	36.3%	29.3%* (22.6, 35.9)	8.3%	44.0%	35.1%* (28.6, 41.6)
Prior biologic failure[+]	1.7%	27.0%	25.3%	4.8%	37.1%	32.3%

Without prior biologic failure+	13.2%	46.8%	33.6%	12.0%	51.2%	39.2%
Histologic-endoscopic mucosal healing[d]	6.6%	30.1%	23.7%* (17.5, 30.0)	5.9%	36.7%	30.1%* (24.1, 36.2)
Prior biologic failure+	1.4%	22.7%	21.3%	4.6%	30.7%	26.1%
Without prior biologic failure+	11.8%	38.2%	26.4%	7.2%	42.9%	35.7%
Deep mucosal healing[e]	1.3%	10.7%	9.7%* (5.7, 13.7)	1.7%	13.5%	11.3%* (7.2, 15.3)
Prior biologic failure+	0	6.5%	6.5%	1.1%	9.2%	8.1%
Without prior biologic failure+	2.6%	15.4%	12.8%	2.4%	17.9%	15.5%

Abbreviations: PBO = placebo; UPA= upadacitinib; aMS = adapted Mayo Score, based on the Mayo Scoring system (excluding Physician's Global Assessment), which ranged from 0 to 9 and has three subscores that were each scored 0 (normal) to 3 (most severe): stool frequency subscore (SFS), rectal bleeding subscore (RBS) and a centrally-reviewed endoscopy subscore (ES).
+The number of “Prior biologic failure” patients in UC-1 and UC-2 are 78 and 89 in the placebo group, and 168 and 173 in the upadacitinib 45 mg group, respectively; the number of “Without prior biologic failure” patients in UC-1 and UC-2 are 76 and 85 in the placebo group, and 151 and 168 in the upadacitinib 45 mg group, respectively.
*p <0.001, adjusted treatment difference (95% CI)
[a] Per aMS: SFS≤ 1 and not greater than baseline, RBS = 0, ES ≤ 1 without friability
[b] Per aMS: decrease ≥ 2 points and ≥ 30% from baseline and a decrease in RBS ≥ 1 from baseline or an absolute RBS ≤ 1.
[c]ES ≤ 1 without friability
[d] ES ≤ 1 without friability and Geboes score ≤ 3.1 (indicating neutrophil infiltration in < 5% of crypts, no crypt destruction, and no erosions, ulcerations, or granulation tissue.)
[e] ES = 0, Geboes score < 2 (indicating no neutrophil in crypts or lamina propria and no increase in eosinophil, no crypt destruction, and no erosions, ulcerations, or granulation tissue)

Disease activity and symptoms
The partial adapted Mayo score (paMS) is composed of SFS and RBS. Symptomatic response per paMS is defined as a decrease of ≥1 point and ≥30% from baseline and a decrease in RBS ≥ 1 or an absolute RBS ≤1. Statistically significant improvement compared to placebo per paMS was seen as early as week 2 (UC-1: 60.1% vs 27.3% and UC-2: 63.3% vs 25.9%).

Extended induction
A total of 125 patients in UC-1 and UC-2 who did not achieve clinical response after 8 weeks of treatment with upadacitinib 45 mg once daily entered an 8-week open-label extended induction period. After the treatment of an additional 8 weeks (16 weeks total) of upadacitinib 45 mg once daily, 48.3% of patients achieved clinical response per aMS. Among patients who responded to treatment of 16-week upadacitinib 45 mg once daily, 35.7% and 66.7% of patients maintained clinical response per aMS and 19.0% and 33.3% of patients achieved clinical remission per aMS at week 52 with maintenance treatment of upadacitinib 15 mg and 30 mg once daily, respectively.

<u>*Maintenance study (UC-3)*</u>
The efficacy analysis for UC-3 was evaluated in 451 patients who achieved clinical response per aMS with 8-week upadacitinib 45 mg once daily induction treatment. Patients were randomised to receive upadacitinib 15 mg, 30 mg or placebo once daily for up to 52 weeks.

The primary endpoint was clinical remission per aMS at week 52. Table 23 shows the key secondary endpoints including maintenance of clinical remission, corticosteroid-free clinical remission, mucosal healing, histologic-endoscopic mucosal healing and deep mucosal healing.

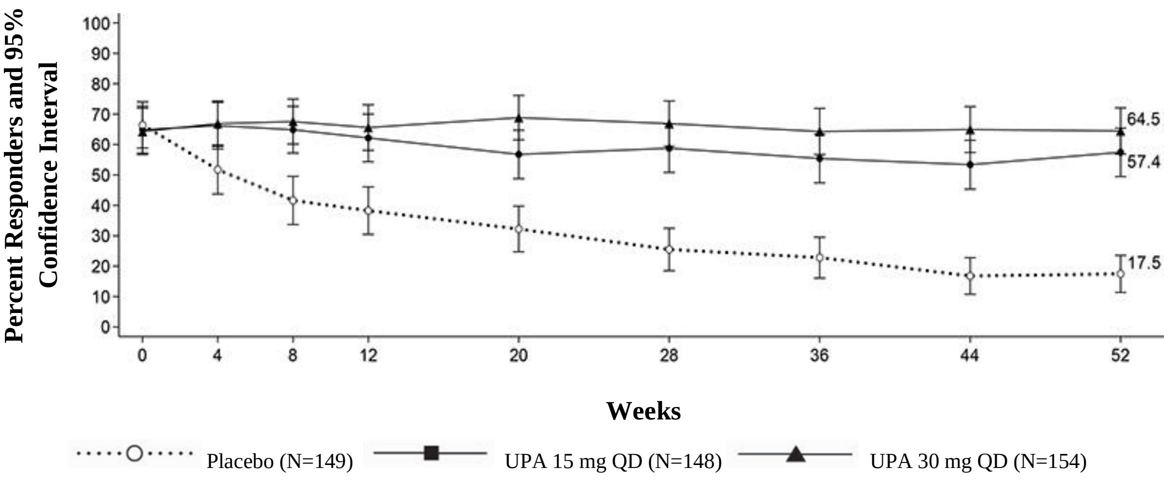
Table 23. Proportion of patients meeting primary and key secondary efficacy endpoints at week 52 in the maintenance study UC-3

	PBO N=149	UPA 15 mg N=148	UPA 30 mg N=154	Treatment Difference 15 mg vs PBO (95% CI)	Treatment Difference 30 mg vs PBO (95% CI)
Clinical remission[a]	12.1%	42.3%	51.7%	30.7%* (21.7, 39.8)	39.0%* (29.7, 48.2)
Prior biologic failure[+]	7.5%	40.5%	49.1%	33.0%	41.6%
Without prior biologic failure[+]	17.6%	43.9%	54.0%	26.3%	36.3%
Maintenance of clinical remission[b]	N = 54 22.2%	N = 47 59.2%	N = 58 69.7%	37.4%* (20.3, 54.6)	47.0%* (30.7, 63.3)
Prior biologic failure	N = 22 13.6%	N = 17 76.5%	N = 20 73.0%	62.8%	59.4%
Without prior biologic failure	N = 32 28.1%	N = 30 49.4%	N = 38 68.0%	21.3%	39.9%
Corticosteroid-free clinical remission[c]	N = 54 22.2%	N = 47 57.1%	N = 58 68.0%	35.4%* (18.2, 52.7)	45.1%* (28.7, 61.6)
Prior biologic failure	N = 22 13.6%	N = 17 70.6%	N = 20 73.0%	57.0%	59.4%
Without prior biologic failure	N = 32 28.1%	N = 30 49.4%	N = 38 65.4%	21.3%	37.2%
Mucosal healing[d]	14.5%	48.7%	61.6%	34.4%* (25.1, 43.7)	46.3%* (36.7, 55.8)
Prior biologic failure[+]	7.8%	43.3%	56.1%	35.5%	48.3%
Without prior biologic failure[+]	22.5%	53.6%	66.6%	31.1%	44.1%
Histologic-endoscopic mucosal healing[e]	11.9%	35.0%	49.8%	23.8%* (14.8, 32.8)	37.3%* (27.8, 46.8)
Prior biologic failure[+]	5.2%	32.9%	47.6%	27.7%	42.4%
Without prior biologic failure[+]	20.0%	36.9%	51.8%	16.9%	31.8%
Deep mucosal healing[f]	4.7%	17.6%	19.0%	13.0%* (6.0, 20.0)	13.6%* (6.6, 20.6)
Prior biologic failure[+]	2.5%	17.2%	16.1%	14.7%	13.6%
Without prior biologic failure[+]	7.5%	18.0%	21.6%	10.6%	14.2%

Abbreviations: PBO = placebo; UPA= upadacitinib; aMS = adapted Mayo Score, based on the Mayo Scoring system (excluding Physician's Global Assessment), which ranged from 0 to 9 and has three subscores that were each scored 0 (normal) to 3 (most severe): stool frequency subscore (SFS), rectal bleeding subscore (RBS) and a centrally-reviewed endoscopy subscore (ES).
[+]The number of "Prior biologic failure" patients are 81, 71, and 73 in the placebo, upadacitinib 15 mg, and 30 mg group, respectively. The number of "Without prior biologic failure" patients are 68, 77, and 81 in the placebo, upadacitinib 15 mg, and 30 mg group, respectively.
* p <0.001, adjusted treatment difference (95% CI)
[a] Per aMS: SFS≤ 1 and not greater than baseline, RBS = 0, ES ≤ 1 without friability
[b] Clinical remission per aMS at Week 52 among patients who achieved clinical remission at the end of induction treatment.
[c] Clinical remission per aMS at Week 52 and corticosteroid-free for ≥90 days immediately preceding Week 52 among patients who achieved clinical remission at the end of the induction treatment.
[d] ES ≤1 without friability
[e] ES ≤1 without friability and Geboes score ≤ 3.1 (indicating neutrophil infiltration in <5% of crypts, no crypt destruction and no erosions, ulcerations or granulation tissue).
[f] ES = 0, Geboes score < 2 (indicating no neutrophil in crypts or lamina propria and no increase in eosinophil, no crypt destruction, and no erosions, ulcerations or granulation tissue).

Disease symptoms
Symptomatic remission per paMS, defined as SFS ≤ 1 and RBS = 0, was achieved over time through week 52 in more patients treated with both upadacitinib 15 mg and 30 mg once daily compared with placebo (Figure 6).

Figure 6 Proportion of patients with symptomatic remission per partial adapted Mayo score over time in maintenance study UC-3



Endoscopic assessment
Endoscopic remission (normalisation of the endoscopic appearance of the mucosa) was defined as ES of 0. At week 8, a significantly greater proportion of patients treated with upadacitinib 45 mg once daily compared to placebo achieved endoscopic remission (UC-1: 13.7% vs 1.3%, UC-2: 18.2% vs 1.7%). In UC-3, a significantly greater proportion of patients treated with upadacitinib 15 mg and 30 mg once daily compared to placebo achieved endoscopic remission at week 52 (24.2% and 25.9% vs 5.6%). Maintenance of mucosal healing at week 52 (ES ≤1 without friability) was seen in a significantly greater proportion of patients treated with upadacitinib 15 mg and 30 mg once daily compared to placebo (61.6% and 69.5% vs 19.2%) among patients who achieved mucosal healing at the end of induction.

Quality of life
Patients treated with upadacitinib demonstrated significantly greater and clinically meaningful improvement in health-related quality of life measured by the Inflammatory Bowel Disease Questionnaire (IBDQ) total score compared to placebo. Improvements were seen in all 4 domain scores: systemic symptoms (including fatigue), social function, emotional function and bowel symptoms (including abdominal pain and bowel urgency). Changes in IBDQ total score at week 8 from baseline with upadacitinib 45 mg once daily compared to placebo were 55.3 and 21.7 in UC-1 and 52.2 and 21.1 in UC-2, respectively. Changes in IBDQ total score at week 52 from baseline were 49.2, 58.9 and 17.9 in patients treated with upadacitinib 15 mg, 30 mg once daily and placebo, respectively.

Long-Term Extension Study (UC-4)
Patients who achieved clinical remission in UC-3 per aMS at 1 year were eligible to continue with the same dose in the extension study (UC-4). After a total of 3 years, 78.6% (55/70) and 84.3% (75/89) of patients maintained clinical remission and 64.7% (22/34) and 74.1% (40/54) of patients maintained endoscopic remission with RINVOQ 15 mg and 30 mg, respectively. Quality of life improvements were also maintained at 3 years. The safety profile of RINVOQ with long-term treatment was consistent with that in the placebo controlled period.

Crohn's disease

The efficacy and safety of upadacitinib was evaluated in three multicenter, double-blind, placebo-controlled Phase 3 studies: two induction studies, CD-1 (U-EXCEED) and CD-2 (U-EXCEL), followed by a 52-week maintenance treatment and long-term extension study, CD-3 (U-ENDURE). The co-primary endpoints were clinical remission and endoscopic response at week 12 for CD-1 and CD-2, and at week 52 for CD-3.

Enrolled patients were 18 to 75 years of age with moderately to severely active Crohn's disease (CD), defined as an average daily very soft or liquid stool frequency (SF) ≥ 4 and/or average daily abdominal pain score (APS) ≥ 2, and a centrally-reviewed Simple Endoscopic Score for CD (SES-CD) of ≥ 6, or ≥ 4 for isolated ileal disease, excluding the narrowing component. Patients with symptomatic bowel strictures were excluded from CD studies.

Induction studies (CD-1 and CD-2)

In CD-1 and CD-2, 1021 patients (495 and 526 patients, respectively) were randomised to upadacitinib 45 mg once daily or placebo for 12 weeks with a 2:1 treatment allocation ratio.

In CD-1, all patients had inadequate response or were intolerant to treatment with one or more biologic therapies (prior biologic failure). Of these patients, 61% (301/495) had inadequate response or were intolerant to two or more biologic therapies.

In CD-2, 45% (239/526) patients had an inadequate response or were intolerant to treatment with one or more biologic therapies (prior biologic failure), and 55% (287/526) had an inadequate response or were intolerant to treatment with conventional therapies but not to biologic therapy (without prior biologic failure).

At baseline in CD-1 and CD-2, 34% and 36% of patients received corticosteroids, 7% and 3% of patients received immunomodulators, and 15% and 25% of patients received aminosalicylates.

In both studies, patients receiving corticosteroids at baseline initiated a corticosteroid taper regimen starting at week 4.

Both studies included a 12-week extended treatment period with upadacitinib 30 mg once daily for patients who received upadacitinib 45 mg once daily and did not achieve clinical response per SF/APS (≥ 30% decrease in average daily very soft or liquid SF and/or ≥ 30% decrease in average daily APS and neither greater than baseline) at week 12.

Clinical disease activity and symptoms

In CD-1 and CD-2, a significantly greater proportion of patients treated with upadacitinib 45 mg achieved the co-primary endpoint of clinical remission at week 12 compared to placebo (Table 24). Onset of efficacy was rapid and achieved as early as week 2 (Table 24).

In both studies, patients receiving upadacitinib 45 mg experienced significantly greater improvement from baseline in fatigue, as measured by FACIT-F score at week 12 compared to placebo.

Endoscopic assessment

In CD-1 and CD-2, a significantly greater proportion of patients treated with upadacitinib 45 mg achieved the co-primary endpoint of endoscopic response at week 12 compared to placebo (Table 24). In CD-1 and CD-2, a greater proportion of patients treated with upadacitinib 45 mg (14% and 19%, respectively) compared to placebo (0% and 5%, respectively) achieved SES-CD 0-2.

Table 24. Proportion of patients meeting primary and additional efficacy endpoints in induction studies CD-1 and CD-2

Study	**CD-1 (U-EXCEED)**			**CD-2 (U-EXCEL)**		
Treatment Group	**PBO N=171**	**UPA 45 mg N=324**	**Treatment Difference (95% CI)**	**PBO N=176**	**UPA 45 mg N=350**	**Treatment Difference (95% CI)**
Co-Primary Endpoints at Week 12						
Clinical remission[a]	14%	40%	26% (19, 33)*	22%	51%	29% (21, 36)*
Prior biologic failure				N=78 14%	N=161 47%	33% (22, 44)
Without prior biologic failure				N=98 29%	N=189 54%	26% (14, 37)
Endoscopic response[b]	4%	35%	31% (25, 37)*	13%	46%	33% (26, 40)*
Prior biologic failure				N=78 9%	N=161 38%	29% (19, 39)
Without prior biologic failure				N=98 16%	N=189 52%	36% (25, 46)
Additional Endpoints at Week 12						
Clinical remission per CDAI[c]	21%	39%	18% (10, 26)*	29%	49%	21% (13, 29)*
Clinical response (CR-100)[d]	27%	51%	23% (14, 31)*	37%	57%	20% (11, 28)*
Corticosteroid-free clinical remission[a,e]	N=60 7%	N=108 37%	30% (19, 41)*	N=64 13%	N=126 44%	33% (22, 44)*
Endoscopic remission[f]	2%	19%	17% (12, 22)*	7%	29%	22% (16, 28)*
Mucosal healing[g]	N=171 0%	N= 322 17%	17% (13, 21)***	N=174 5%	N=349 25%	20% (14, 25)***
Early Onset Endpoints						
Clinical remission at Week 4[a]	9%	32%	23% (17, 30)*	15%	36%	21% (14, 28)*
CR-100 at Week 2[d]	12%	33%	21% (14, 28)*	20%	32%	12% (4, 19)**
Abbreviation: PBO = placebo, UPA = upadacitinib						

* p < 0.001, adjusted treatment difference (95% CI) ** p < 0.01, adjusted treatment difference (95% CI) *** nominal p < 0.001 UPA vs PBO comparison, adjusted treatment difference (95% CI) a Average daily SF ≤ 2.8 and APS ≤ 1.0 and neither greater than baseline b Decrease in SES-CD > 50% from baseline of the induction study (or for patients with an SES-CD of 4 at baseline of the induction study, at least a 2-point reduction from baseline of the induction study) c CDAI < 150 d Decrease of at least 100 points in CDAI from baseline e Discontinuation of steroid and achievement of clinical remission among patients on steroid at baseline f SES-CD ≤ 4 and at least a 2-point reduction versus baseline and no subscore > 1 in any individual variable g SES-CD ulcerated surface subscore of 0 in patients with SES-CD ulcerated surface subscore ≥ 1 at baseline

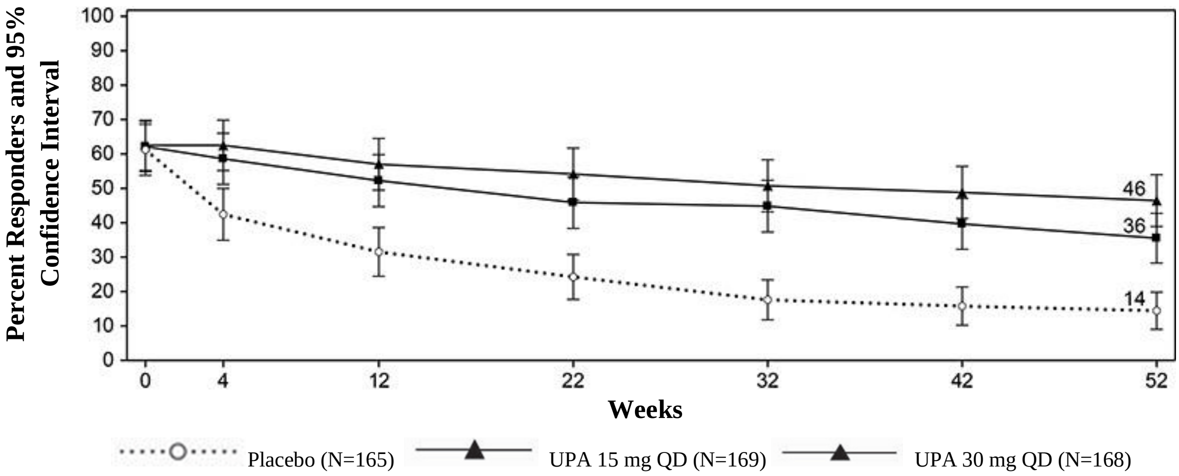
<u>*Maintenance study (CD-3)*</u>

The efficacy analysis for CD-3 evaluated 502 patients who achieved clinical response per SF/APS with the 12-week upadacitinib 45 mg once daily induction treatment. Patients were re-randomised to receive a maintenance regimen of either upadacitinib 15 mg or 30 mg once daily or placebo for 52 weeks.

Clinical disease activity and symptoms

A significantly greater proportion of patients treated with upadacitinib 15 mg and 30 mg achieved the co-primary endpoint of clinical remission at week 52 compared to placebo (Figure 7, Table 25).

Figure 7 Proportion of patients achieving clinical remission in maintenance study CD-3



Patients receiving upadacitinib 30 mg experienced significantly greater improvement from baseline in fatigue, as measured by FACIT-F score at week 52 compared to placebo.

Table 25. Proportion of patients meeting primary and additional efficacy endpoints at week 52 in maintenance study CD-3

Treatment Group	PBO[+] N=165	UPA 15 mg N=169	UPA 30 mg N=168	Treatment Difference 15 mg vs PBO (95% CI)	Treatment Difference 30 mg vs PBO (95% CI)
Co-Primary Endpoints					
Clinical remission[a]	14%	36%	46%	22% (14, 30)[*]	32% (23, 40)[*]
Prior biologic failure	N=126 9%	N=124 32%	N=127 43%	24% (14, 33)	34% (24, 44)
Without prior biologic failure	N=39 33%	N=45 44%	N=41 59%	12% (-9, 33)	26% (5, 47)
Endoscopic response[b]	7%	28%	40%	21% (14, 28)[*]	34% (26, 41)[*]
Prior biologic failure	N=126 4%	N=124 23%	N=127 39%	19% (11, 27)	35% (26, 44)
Without prior biologic failure	N=39 18%	N=45 40%	N=41 44%	22% (3, 41)	26% (7, 45)
Additional Endpoints					
Clinical remission per CDAI[c]	15%	37%	48%	24% (15, 32)[*]	33% (24, 42)[*]
Clinical response (CR-100)[d]	15%	41%	51%	27% (18, 36)[*]	36% (28, 45)[*]
Corticosteroid-free clinical remission[a,e]	14%	35%	45%	21% (13, 30)[*]	30% (21, 39)[*]
Maintenance of clinical remission [a,f]	N=101 20%	N=105 50%	N=105 60%	32% (20, 44)[*]	40% (28, 52)[*]
Endoscopic remission[g]	5%	19%	29%	14% (8, 21)[*]	24% (16, 31)[*]
Mucosal healing[h]	N=164 4%	N=167 13%	N=168 24%	10% (4, 16)[***]	21% (14, 27)[***]
Deep remission [a,i]	4%	14%	23%	10% (4, 16)[**]	18% (11, 25)[*]

Abbreviation: PBO = placebo, UPA = upadacitinib

[+] The placebo group consisted of patients who achieved clinical response per SF/APS with upadacitinib 45 mg at the end of the induction study and were randomised to receive placebo at the start of maintenance therapy

[*] $p < 0.001$, adjusted treatment difference (95% CI)

[**] $p < 0.01$, adjusted treatment difference (95% CI)

[***] nominal $p < 0.001$ UPA vs PBO comparison, adjusted treatment difference (95% CI)

[a] Average daily SF ≤ 2.8 and APS ≤ 1.0 and neither greater than baseline

[b] Decrease in SES-CD > 50% from baseline of the induction study (or for patients with an SES-CD of 4 at baseline of the induction study, at least a 2-point reduction from baseline of the induction study)

[c] CDAI < 150

[d] Reduction of CDAI ≥ 100 points from baseline

[e] Corticosteroid-free for 90 days prior to week 52 and achievement of clinical remission. Among the subset of patients who were on corticosteroids at induction baseline, 38% (N=63) in upadacitinib

15 mg group, 38% (N=63) in upadacitinib 30 mg group, and 5% (N=61) in placebo were corticosteroid-free for 90 days prior to week 52 and in clinical remission [f] Defined as achievement of clinical remission at Week 52 in patients who achieved clinical remission at the entry of the maintenance study [g] SES-CD ≤ 4 and at least a 2-point reduction versus baseline and no subscore >1 in any individual variable [h] SES-CD ulcerated surface subscore of 0 in patients with SES-CD ulcerated surface subscore ≥ 1 at baseline [i] Clinical remission and endoscopic remission

Patients who were not in clinical response per SF/APS to upadacitinib induction at week 12 in CD-1 and CD-2 (122 patients) received upadacitinib 30 mg once daily for an additional 12 weeks. Of these patients, 53% achieved clinical response at week 24. Of the patients who responded to the extended treatment period and continued to receive maintenance treatment with upadacitinib 30 mg, 25% achieved clinical remission and 22% achieved endoscopic response at week 52.

Endoscopic assessment

In CD-3, a significantly greater proportion of patients treated with upadacitinib 15 mg and 30 mg achieved the co-primary endpoint of endoscopic response at week 52 compared to placebo (Table 25). In addition to the endoscopic endpoints described in Table 23, a greater proportion of patients treated with upadacitinib 15 mg and 30 mg (11% and 21%, respectively) compared to placebo (3%) achieved SES-CD 0-2 at week 52. Corticosteroid-free endoscopic remission among patients on steroid at baseline was achieved in a greater proportion of patients treated with upadacitinib 15 mg and 30 mg (17% and 25%, respectively) compared to placebo (3%) at week 52.

Resolution of extra-intestinal manifestations

Resolution of extra-intestinal manifestations was observed in a greater proportion of patients treated with upadacitinib 15 mg (25%) and a significantly greater proportion of patients treated with upadacitinib 30 mg (36%) compared to placebo (15%) at week 52.

Rescue treatment

In CD-3, patients who demonstrated inadequate response or lost response during maintenance were eligible to receive rescue treatment with upadacitinib 30 mg. Of the patients who were randomised to upadacitinib 15 mg group and received rescue treatment of upadacitinib 30 mg for at least 12 weeks, 84% (76/90) achieved clinical response per SF/APS and 48% (43/90) achieved clinical remission 12 weeks after initiating rescue.

Health-related quality of life outcomes

Patients treated with upadacitinib achieved greater improvement in health-related quality of life (HRQOL) measured by the Inflammatory Bowel Disease Questionnaire (IBDQ) total score compared to placebo. Improvements were seen in all 4 domain scores: systemic symptoms (including fatigue) and bowel symptoms (including abdominal pain and bowel urgency), as well as social and emotional functioning. Changes from baseline in IBDQ total score at week 12 with upadacitinib 45 mg once daily compared to placebo were 46.0 and 21.6 in CD-1 and 46.3 and 24.4 in CD-2, respectively. Changes in IBDQ total score at week 52 from baseline were 59.3, 64.5 and 46.4 in patients treated with upadacitinib 15 mg, 30 mg once daily and placebo, respectively.

Pharmacokinetic properties

Upadacitinib plasma exposures are proportional to dose over the therapeutic dose range. Steady-state plasma concentrations are achieved within 4 days with minimal accumulation after multiple once-daily administrations.

Absorption

Following oral administration of upadacitinib extended-release formulation, upadacitinib is absorbed with a median T_{max} of 2 to 4 hours. Coadministration of upadacitinib with a high-fat meal had no clinically relevant effect on upadacitinib exposures (increased AUC by 29% and C_{max} by 39% to 60%). In clinical trials, upadacitinib was administered without regard to meals (see **Posology and method of administration** section). *In vitro*, upadacitinib is a substrate for the efflux transporters P-gp and BCRP.

Distribution

Upadacitinib is 52% bound to plasma proteins. Upadacitinib partitions similarly between plasma and blood cellular components, as indicated by the blood to plasma ratio of 1.0.

Metabolism

Upadacitinib metabolism is mediated by CYP3A4 with a potential minor contribution from CYP2D6. The pharmacologic activity of upadacitinib is attributed to the parent molecule. In a human radiolabeled study, unchanged upadacitinib accounted for 79% of the total radioactivity in plasma while the main metabolite (product of monooxidation followed by glucuronidation) accounted for 13% of the total plasma radioactivity. No active metabolites have been identified for upadacitinib.

Elimination

Following single dose administration of [^{14}C]-upadacitinib immediate-release solution, upadacitinib was eliminated predominantly as the unchanged parent substance in urine (24%) and faeces (38%). Approximately 34% of upadacitinib dose was excreted as metabolites. Upadacitinib mean terminal elimination half-life ranged from 9 to 14 hours.

<u>Special populations</u>

<u>Renal impairment</u>

Upadacitinib AUC was 18%, 33%, and 44% higher in subjects with mild (estimated glomerular filtration rate 60 to 89 mL/min/1.73 m^2), moderate (estimated glomerular filtration rate 30 to 59 mL/min/1.73 m^2), and severe (estimated glomerular filtration rate 15 to 29 mL/min/1.73 m^2) renal impairment, respectively, compared to subjects with normal renal function. Upadacitinib C_{max} was similar in subjects with normal and impaired renal function. Mild or moderate renal impairment has no clinically relevant effect on upadacitinib exposure (see **Posology and method of administration** section).

<u>Hepatic impairment</u>

Mild (Child-Pugh A) and moderate (Child-Pugh B) hepatic impairment has no clinically relevant effect on upadacitinib exposure. Upadacitinib AUC was 28% and 24% higher in subjects with mild and moderate hepatic impairment, respectively, compared to subjects with normal liver function. Upadacitinib C_{max} was unchanged in subjects with mild hepatic impairment and 43% higher in subjects with moderate hepatic impairment compared to subjects with normal liver function. Upadacitinib was not studied in patients with severe (Child-Pugh C) hepatic impairment.

Paediatric population

The pharmacokinetics of upadacitinib have not yet been evaluated in a paediatric patients with rheumatoid arthritis,psoriatic arthritis, ankylosing spondylitis, ulcerative colitis and Crohn's disease. (see **Posology and method of administration** section).

Upadacitinib pharmacokinetics and steady-state concentrations are similar for adults and adolescents 12 to 17 years of age with atopic dermatitis.

The pharmacokinetics of upadacitinib in paediatric patients (< 12 years of age) with atopic dermatitis have not been established.

Intrinsic factors

Age, sex, body weight, race, and ethnicity did not have a clinically meaningful effect on upadacitinib exposure. Upadacitinib pharmacokinetics are consistent across patients with rheumatoid arthritis, psoriatic arthritis, non-radiographic axial spondyloarthritis, ankylosing spondylitis, giant cell arteritis, atopic dermatitis, ulcerative colitis and Crohn's disease.

Preclinical safety data

Non-clinical data reveal no special hazard for humans based on conventional studies of safety pharmacology.

Upadacitinib, at exposures (based on AUC) approximately 4 and 10 times the clinical dose of 15 mg, 2 and 5 times the clinical dose of 30 mg, and 1.7 and 4 times the clinical dose of 45 mg in male and female Sprague-Dawley rats, respectively, was not carcinogenic in a 2-year carcinogenicity study in Sprague-Dawley rats. Upadacitinib was not carcinogenic in a 26week carcinogenicity study in CByB6F1-Tg(-HRAS)2Jic transgenic mice.

Upadacitinib was not mutagenic or genotoxic based on the results of *in vitro* and *in vivo* tests for gene mutations and chromosomal aberrations.

Upadacitinib had no effect on fertility in male or female rats at exposures up to approximately 17and 34 times the maximum recommended human dose (MRHD) of 45 mg in males and females, respectively, on an AUC basis in a fertility and early embryonic development study. Doserelated increases in foetal resorptions associated with post-implantation losses in this fertility study in rats were attributed to the developmental/teratogenic effects of upadacitinib. No adverse effects were observed at exposures below clinical exposure (based on AUC). Postimplantation losses were observed at exposures 9 times the clinical exposure at the MRHD of 45 mg (based on AUC).

In animal embryo-foetal development studies, upadacitinib was teratogenic in both rats and rabbits. Upadacitinib resulted in increases in skeletal malformations in rats at 1.6, 0.8 times and 0.6 the clinical exposure (AUC-based) at the 15, 30 and 45 mg (MRHD) doses, respectively. In rabbits an increased incidence of cardiovascular malformations was observed at 15, 7.6 and 5.6 times the clinical exposure at the 15, 30 and 45 mg doses (AUC-based), respectively.

Following administration of upadacitinib to lactating rats, the concentrations of upadacitinib in milk over time generally paralleled those in plasma, with approximately 30-fold higher exposure in milk relative to maternal plasma. Approximately 97% of upadacitinib-related material in milk was the parent molecule, upadacitinib.

PHARMACEUTICAL PARTICULARS

List of Excipients

Tablet contents:

Microcrystalline cellulose
Hypromellose
Mannitol
Tartaric acid
Silica, colloidal anhydrous
Magnesium stearate

Film coating:

Poly(vinyl alcohol)
Macrogol
Talc
Titanium dioxide (E171)
Iron oxide black (E172) (15 mg strength only)
Iron oxide red (E172)
Iron oxide yellow (E172) (45 mg strength only)

Incompatibilities

Not applicable.

Special precautions for storage

Store at or below 30˚C.

Store in the original blister in order to protect from moisture.

Nature and contents of container

Polyvinylchloride/polyethylene/polychlorotrifluoroethylene - aluminium calendar blisters in packs containing 28 extended-release tablets.

Not all pack sizes may be marketed.

Special precautions for disposal

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

Manufacturer:

AbbVie Ireland NL B.V.
Manorhamilton Road
Sligo, Ireland

DATE OF REVISION OF THE TEXT

27 Oct 2025 (CCDS v18)